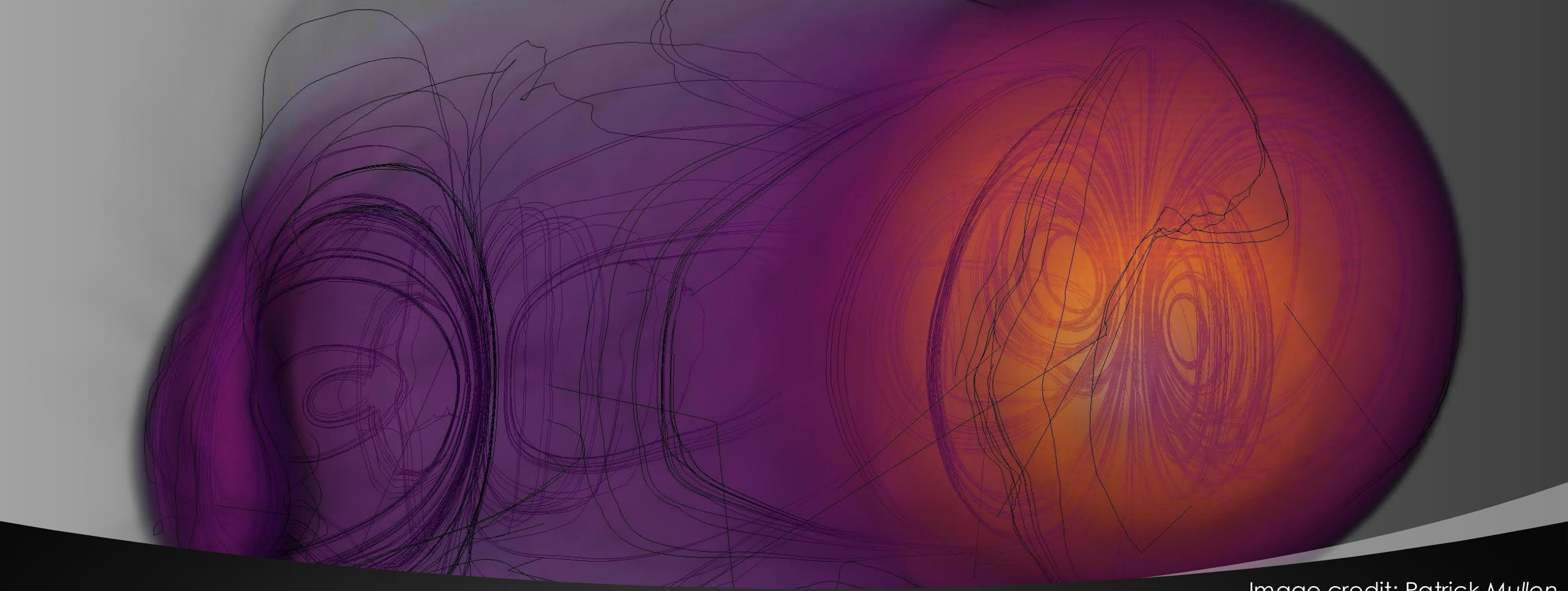


Image credit: Patrick *Mullen*

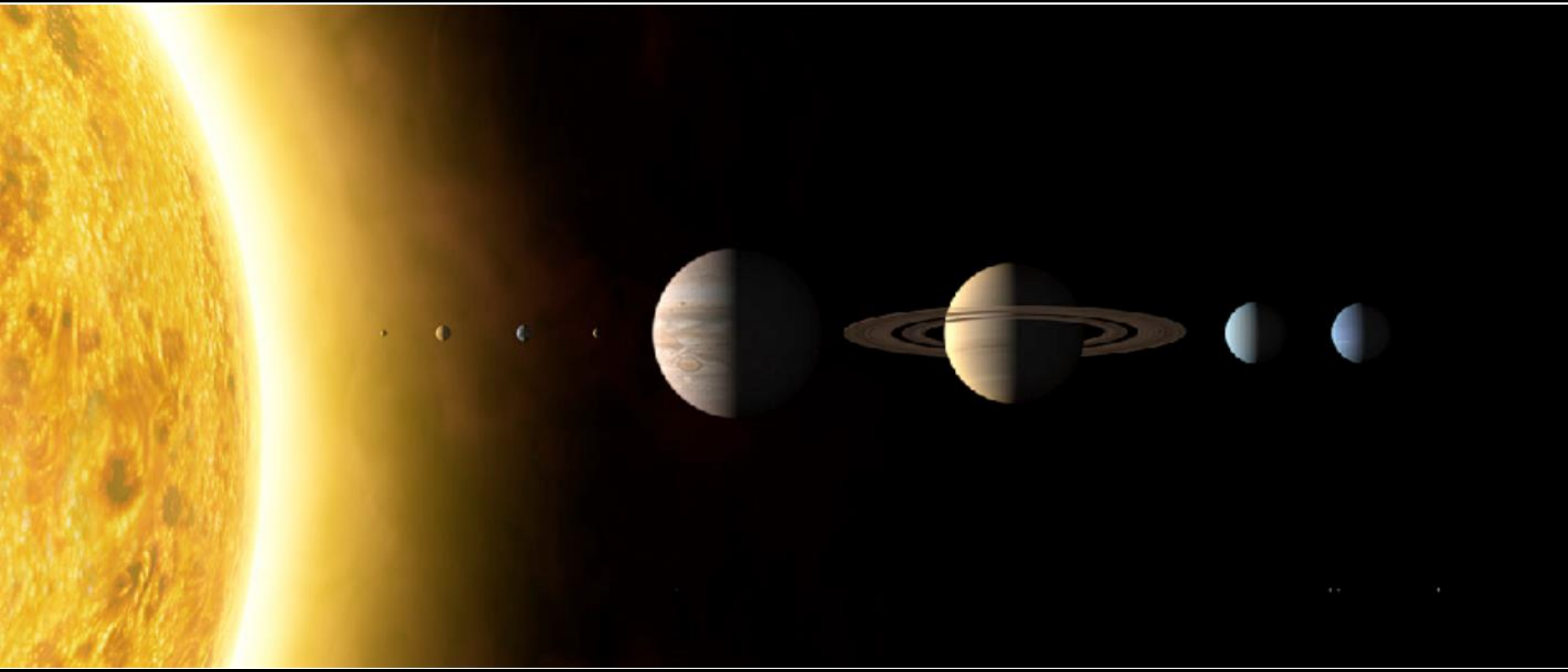
The role of magnetic fields during giant impacts in the solar system

Robert Wissing

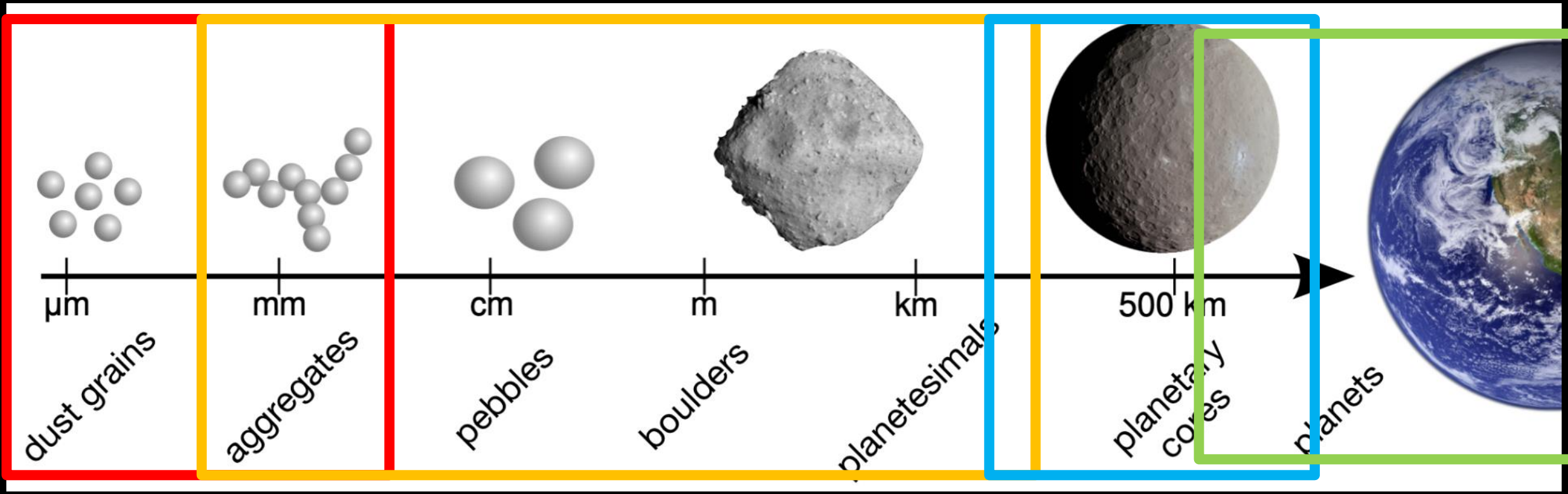
Institute of Theoretical Astrophysics, University of Oslo







From dust to planet



Credit: Joanna Drazkowska

Planetary Stages:

1: Dust coagulation

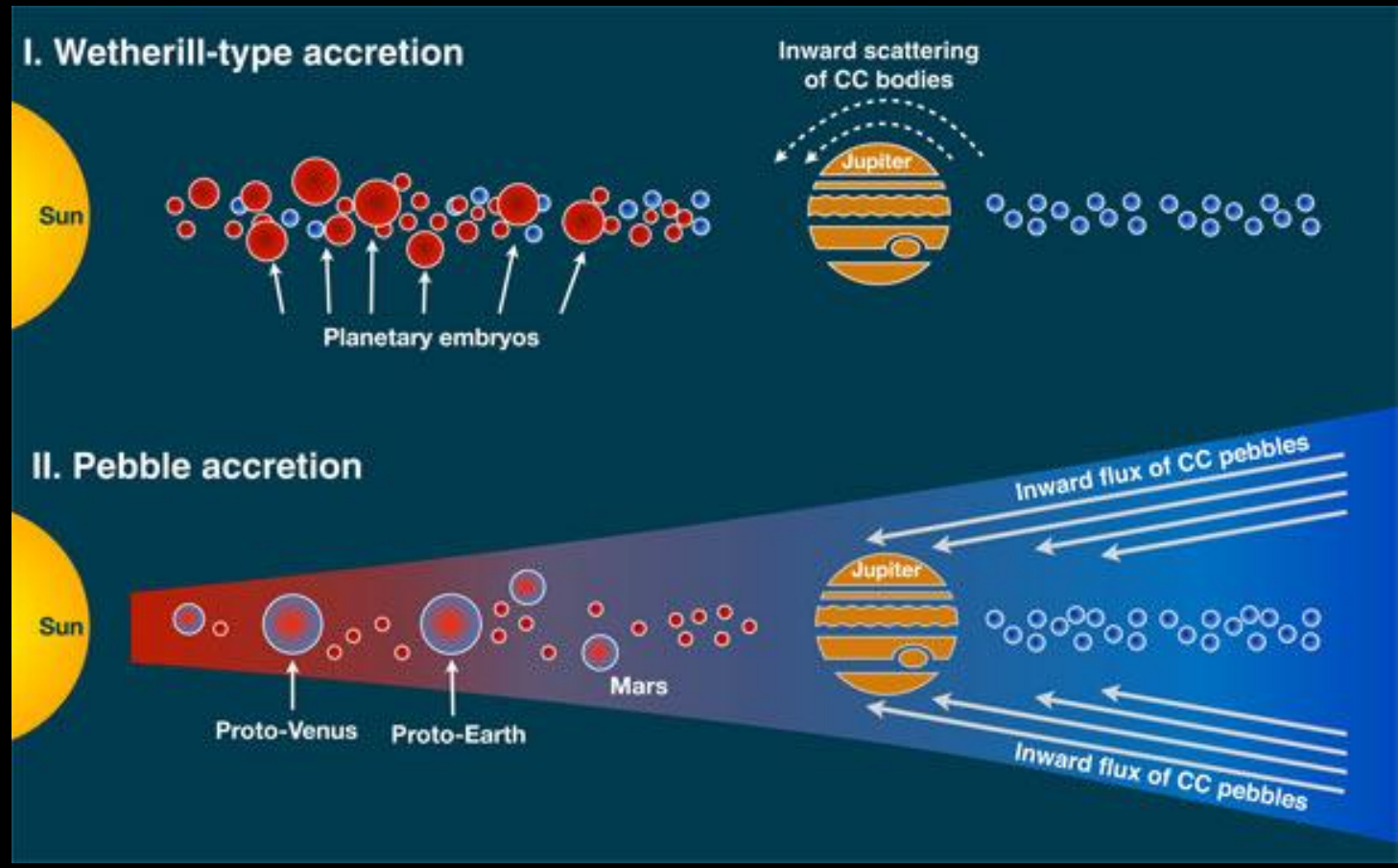
2: Streaming instability

3: Pebble accretion

4: Giant impacts

Terrestrial Planet formation

- Classic Wetherill-type model:
  - Giant impacts among Moon- to Mars-sized planetary embryos after the gas disk dissipated and accreted
- Effective pebble accretion:
  - Formed within the lifetime of the gas disk by efficiently accreting 'pebbles' from the outer Solar System, which drift sunward through the disk due to gas drag.



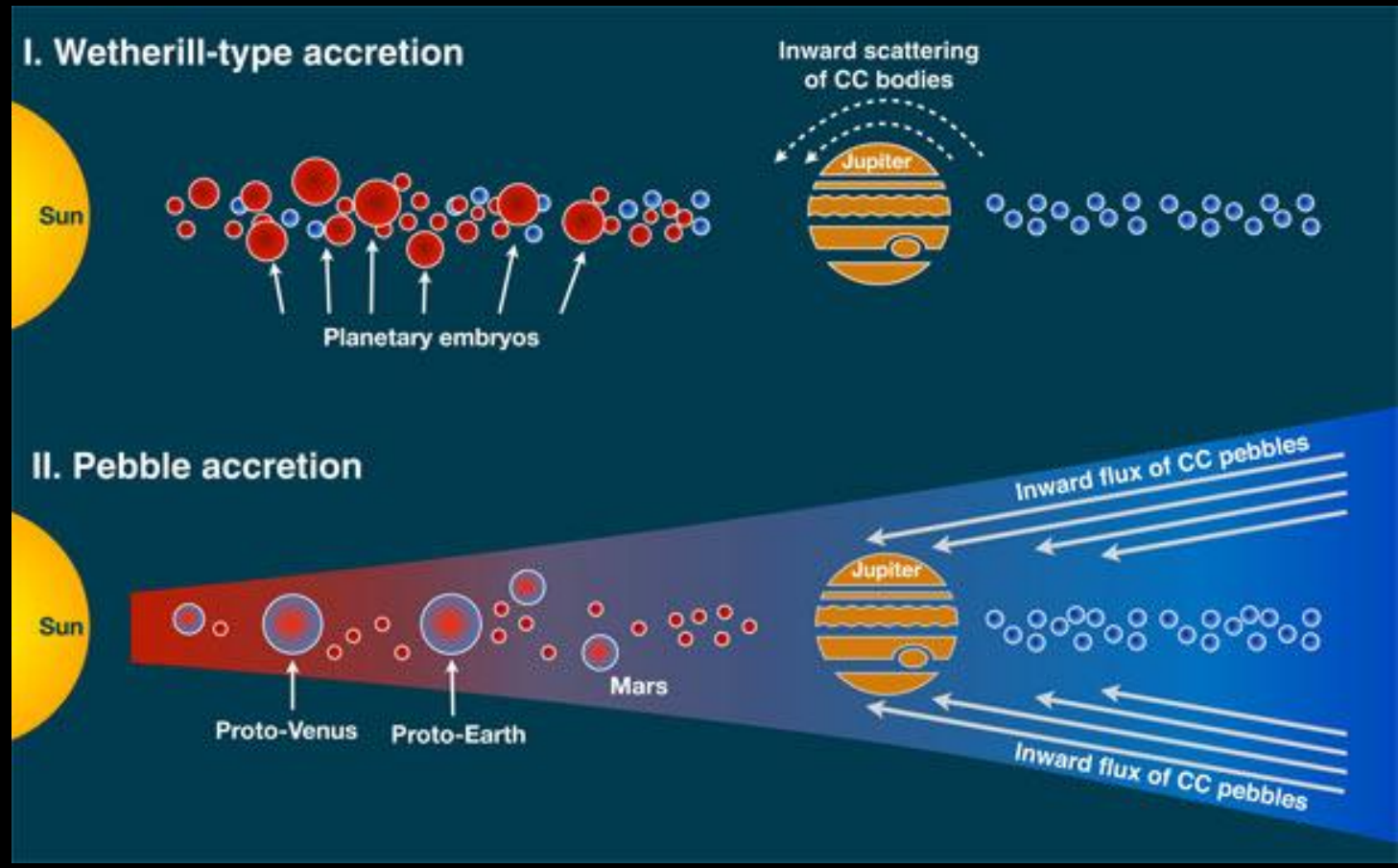
Carbonaceous chondrite (CC)

Burkhardt 2021 et al.

Terrestrial Planet formation

- Classic Wetherill-type model:
  - Giant impacts among Moon- to Mars-sized planetary embryos after the gas disk dissipated and accreted
- Effective pebble accretion:
  - Formed within the lifetime of the gas disk by efficiently accreting 'pebbles' from the outer Solar System, which drift sunward through the disk due to gas drag.

- **Both formation scenarios require a giant impact to form the Moon**

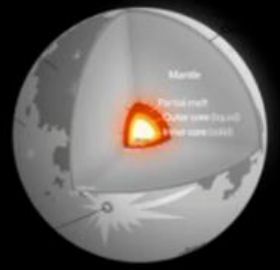


Carbonaceous chondrite (CC)

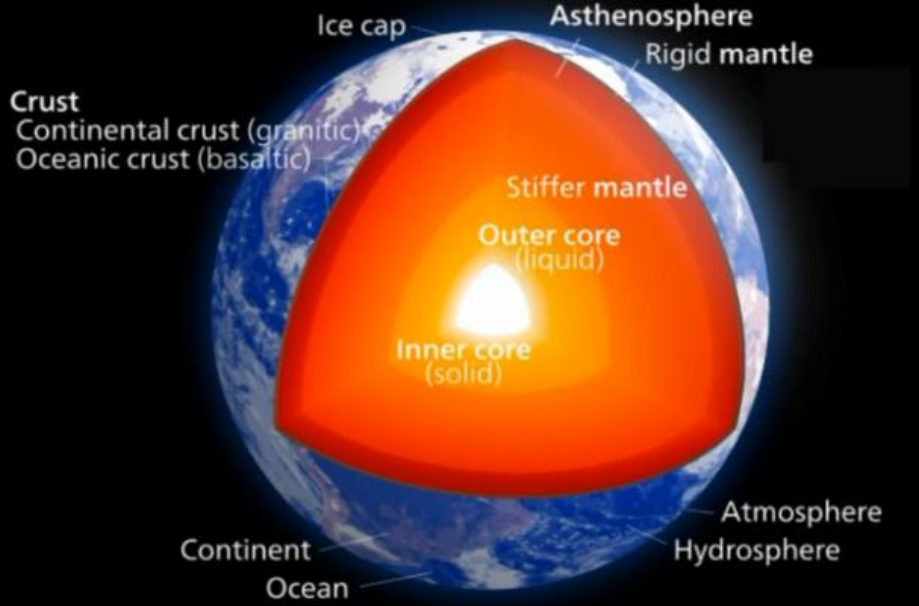
Burkhardt 2021 et al.

Giant impact formed the Moon?





< 10% Iron



~ 30% Iron

- Theory needs to explain the observation of:
  - The mass of the Moon (1% Earth mass)
  - Small iron core in the moon (<10% iron) vs Earth (~30% iron)





- Theory needs to explain the observation of:
  - The mass of the Moon (1% Earth mass)
  - Small iron core in the moon (<10% iron) vs Earth (~30% iron)
  - Formation time predicted to be around 60-100 Myr after birth of solar system
  - Lack of volatiles (water, potassium) on the moon (indicate high temperature formation)

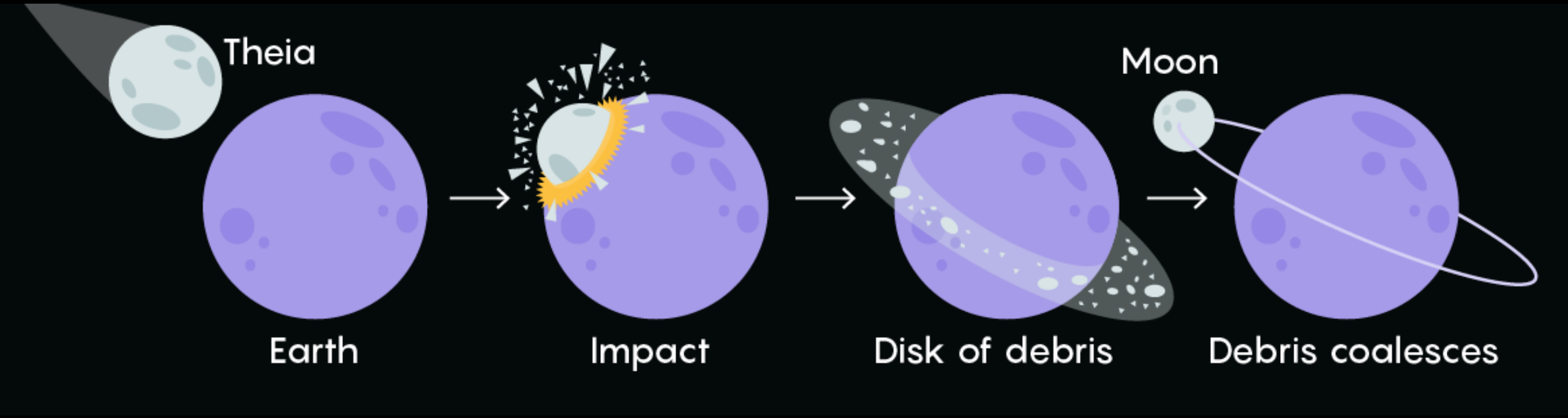


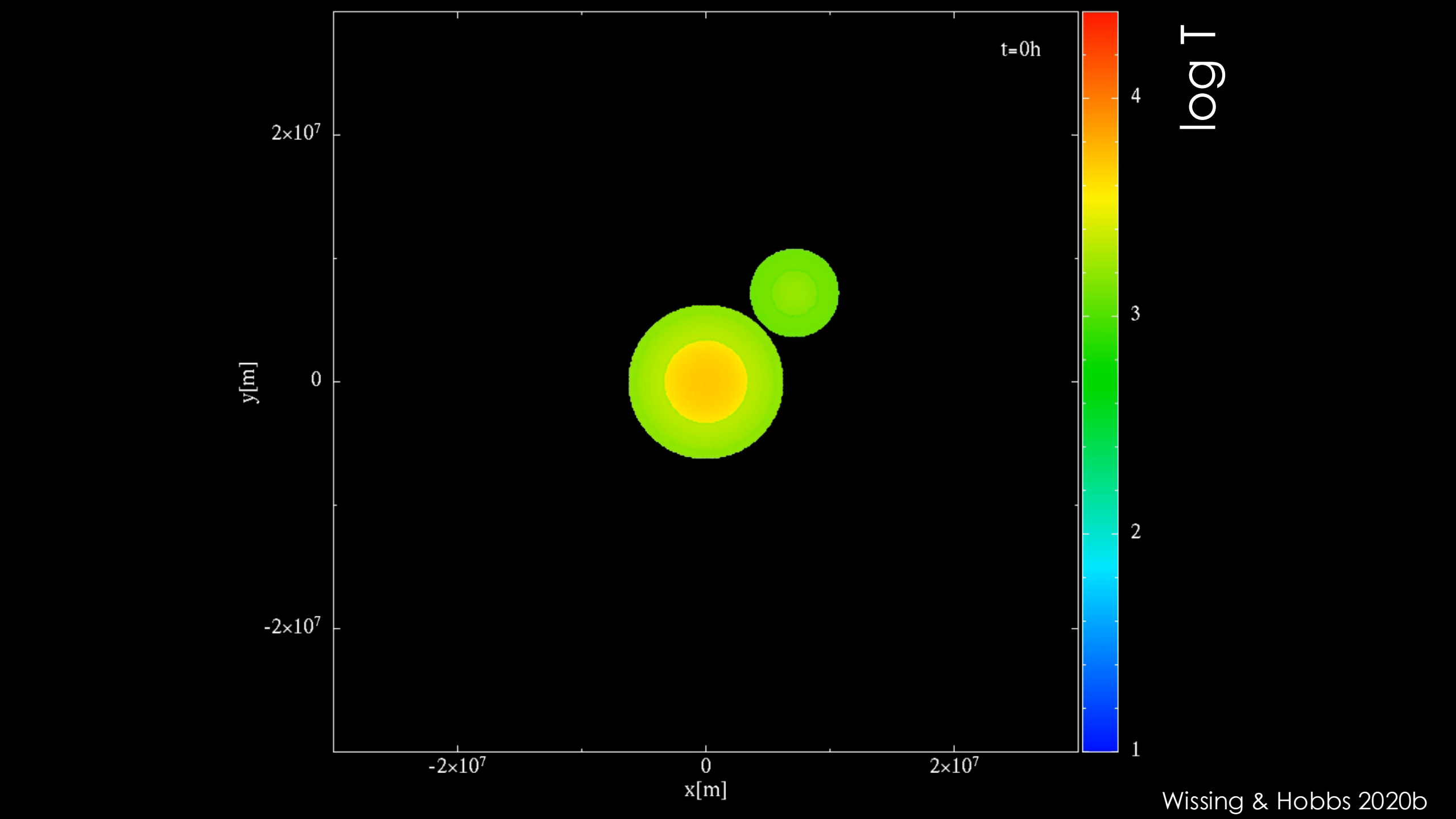
Image credit: Robin Canup



- Theory needs to explain the observation of:
  - The mass of the Moon (1% Earth mass)
  - Small iron core in the moon (<10% iron) vs Earth (~30% iron)
  - Formation time predicted to be around 60-100 Myr after birth of solar system
  - Lack of volatiles (water, potassium) on the moon (indicate high temperature formation)
  - Current earth-moon angular momentum implies a young rapid spinning Earth (5-hour day)

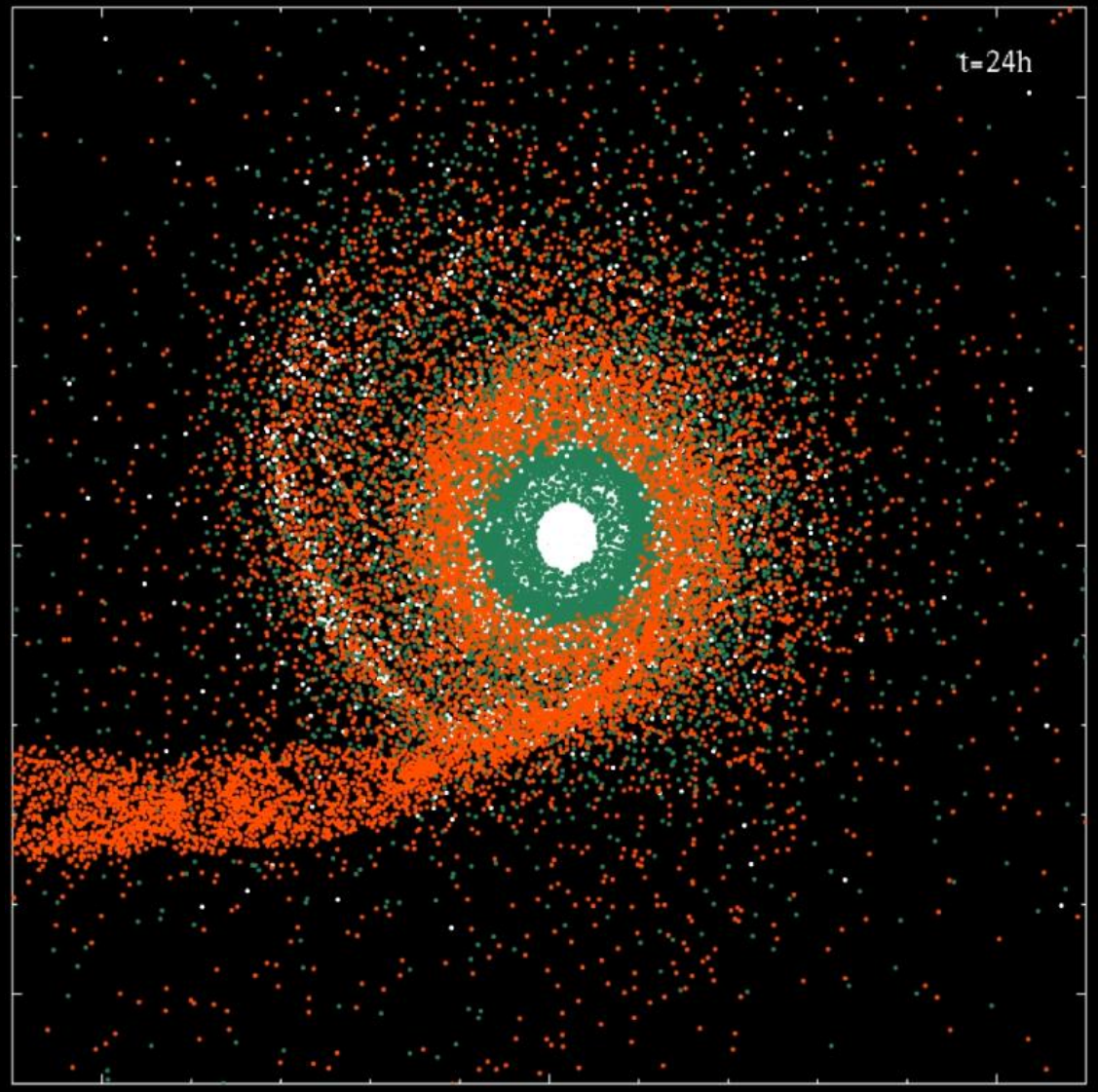
The ”classic” giant impact theory





Wissing & Hobbs 2020b

The "classic" giant impact theory



Wissing & Hobbs 2020b

Iron

$\text{Silicates}_{\text{Target}}$

$\text{Silicates}_{\text{Impactor}}$

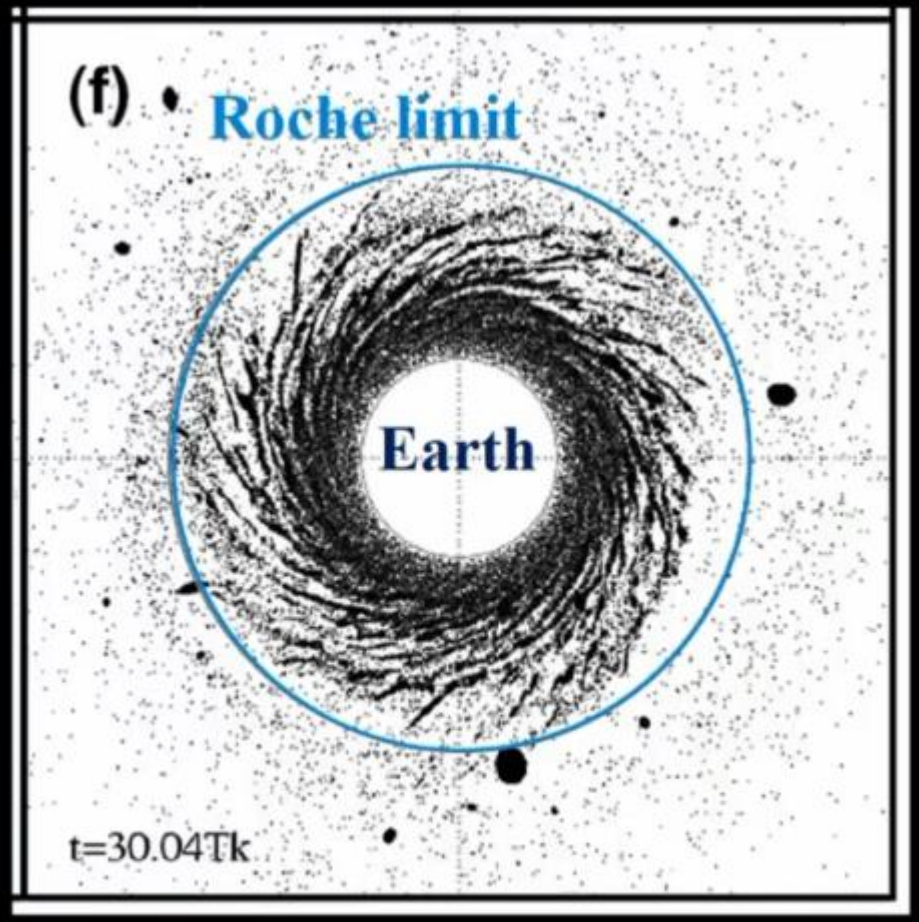
$$L_{sys} \approx L_{EM}$$

$$M_{Disk} \approx 2M_{moon}$$

$$\delta_{FE} \approx 5\%$$

$$\delta_{ft} \approx -50\%$$

The "classic" giant impact theory

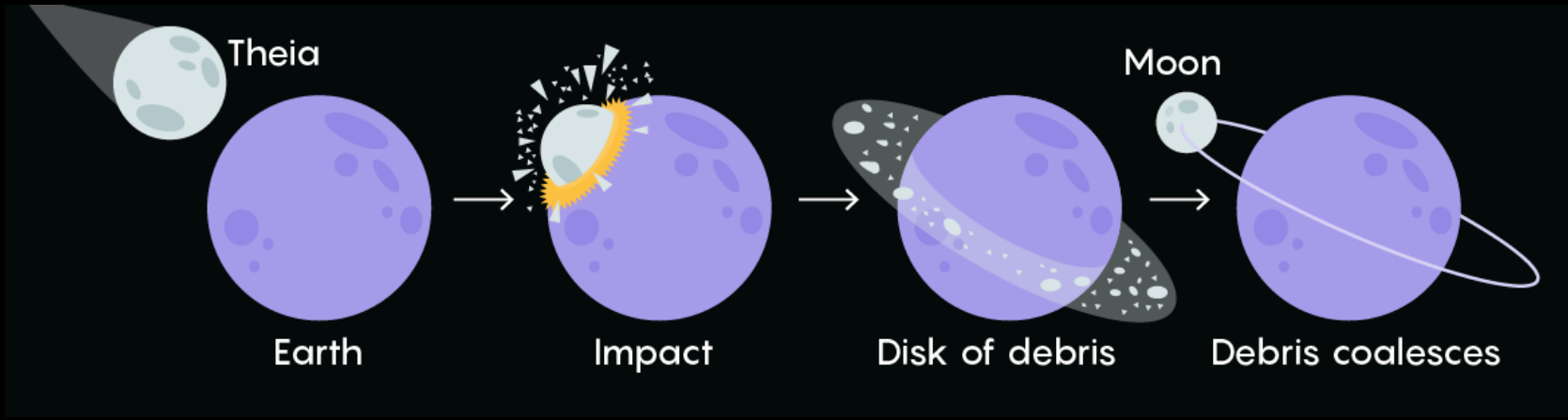


10^5-particle N-body simulation by Takeda & Ida (2001)

- Silicate melt-vapor disk
- Interior to Roche limit:
  - Clumps sheared apart by gravitational tides
  - Disk spreads radially on timescale $t_{spread} \sim 100\ years$ (Takeda and Ida, 2001; Salmon and Canup, 2012;)
- Exterior to Roche limit:
  - Clumps stable and form moonlets
  - Material coalesce quickly to single satellite (~ 1 year) (Salmon and Canup, 2012;)
- Mass of moon after giant impact simulation based on scaling law from disk simulation (Ida et al., 1997; Kokubo et al., 2010)

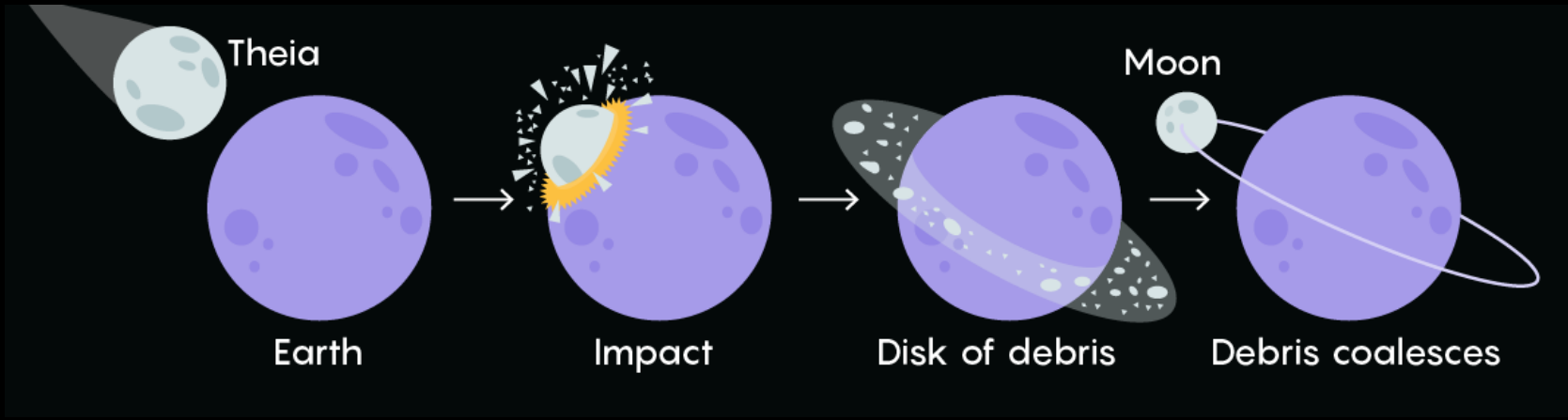
$$M_m \approx M_{moon}$$

The "classic" giant impact theory



- Theory needs to explain the observation of:
  - The mass of the Moon (1% Earth mass)
  - Small iron core in the moon (<10% iron) vs Earth (~30% iron)
  - Formation time predicted to be around 60-100 Myr after birth of solar system
  - Lack of volatiles (water, potassium) on the moon (indicate high temperature formation)
  - Current earth-moon angular momentum implies a young rapid spinning Earth (5-hour day)

The ”classic” giant impact theory



- Theory needs to explain the observation of:
  - The mass of the Moon (1% Earth mass)
  - Small iron core in the moon (<10% iron) vs Earth (~30% iron)
  - Formation time predicted to be around 60-100 Myr after birth of solar system
  - Lack of volatiles (water, potassium) on the moon (indicate high temperature formation)
  - Current earth-moon angular momentum implies a young rapid spinning Earth (5-hour day)
- Modern measurements of lunar rocks reveals near identical isotopic composition!!

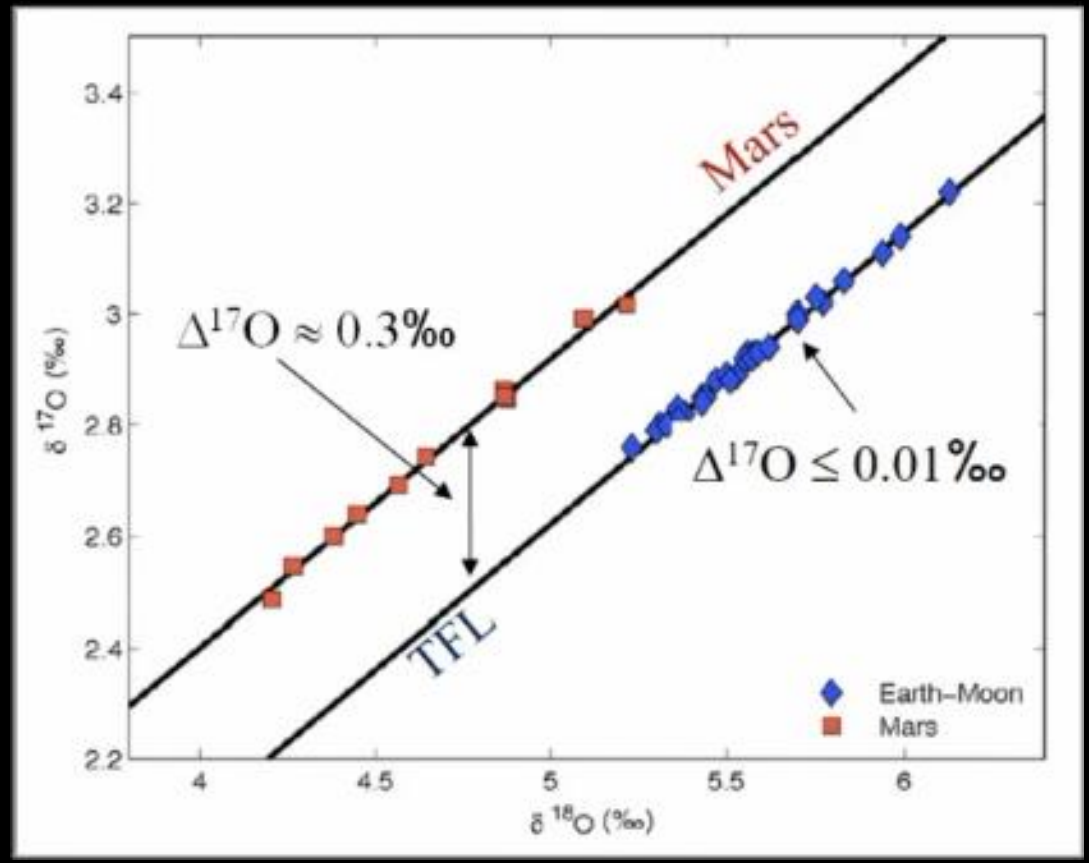
The isotopic crisis

Three stable isotopes of oxygen: ^{16}O, ^{17}O and ^{18}O

$$\delta^{17,18}O = \left[\frac{(^{17,18}O/^{16}O)_{sample}}{(^{17,18}O/^{16}O)_{ref}} - 1 \right] \times 1000$$

- Canonical model predict that moon is made primarily of impactor material!
- Hard to explain if proto-Earth and Theia was formed at different regions/times in the solar system
- Need significant mixing during or after the giant impact

- Oxygen (O) experience large variations in the proto-planetary disk



Pahlevan & Stevenson (2007)

The isotopic crisis

Three stable isotopes of oxygen: ^{16}O, ^{17}O and ^{18}O

$$\delta^{17,18}O = \left[\frac{(^{17,18}O/^{16}O)_{sample}}{(^{17,18}O/^{16}O)_{ref}} - 1 \right] \times 1000$$

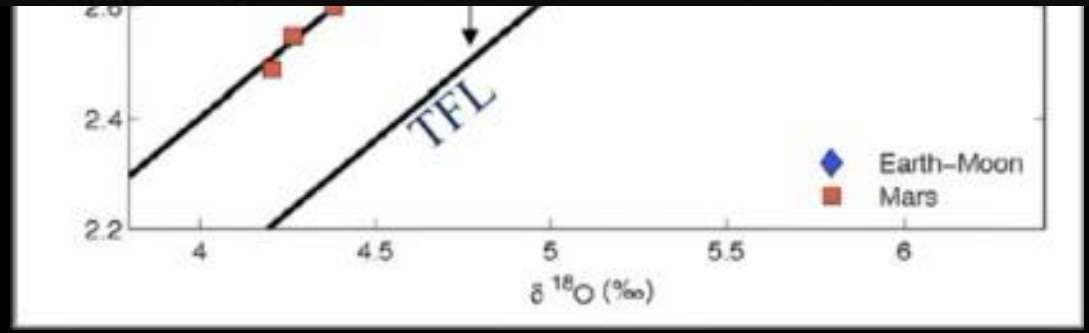
Where does the Moon come from? Doubts on the Giant Impact Hypothesis, *Science News* 2011

Question over theory of lunar formation, *Nature* 2012

Further doubts cast over lunar formation models, *physicsworld.com* 2012

Titanium Casts Doubts on Moon's Origin Story, *Big Ten Science* 2012

- Oxygen (O) experience large variations in the proto-planetary disk
- Tungsten (W) involved in core formation



Pahlevan & Stevenson (2007)

Possible scenarios to resolve crisis

Highly unlikely due to solar system formation history

- ***Scenario 1: Canonical impact + Earth-like composition impactor***

Possible scenarios to resolve crisis

Highly unlikely due to solar system formation history

- ***Scenario 1: Canonical impact + Earth-like composition impactor***
- ***Scenario 2: Hit-and-run impact.***
- ***Scenario 3: Fast-spinning Earth impact***
- ***Scenario 4: Half-Earth impact.***
- ***Scenario 5: General synestia impact with equilibration***

Still impact-rich disk

Need extraction of angular momentum from system $L_{sys} \approx 1.4 L_{EM}$

Need significant extraction of angular momentum from system $L_{sys} \approx 2.5 L_{EM}$

Possible scenarios to resolve crisis

Highly unlikely due to solar system formation history

- *Scenario 1: Canonical impact + Earth-like composition impactor*
- *Scenario 2: Hit-and-run impact.*
- *Scenario 3: Fast-spinning Earth impact*
- *Scenario 4: Half-Earth impact.*
- *Scenario 5: General synestia impact with equilibration*
- *Scenario 6: Multiple impacts.*
- *Scenario 7: Canonical impact + Earth magma ocean*

Still impact-rich disk

Need extraction of angular momentum from system $L_{sys} \approx 1.4 L_{EM}$

Need significant extraction of angular momentum from system $L_{sys} \approx 2.5 L_{EM}$

Dependent on impact history

High "success" rate associated with higher velocity impactors (unlikely)

Equation of state and numerical method needs to be further validated

Initial depth of magma ocean uncertain

Possible scenarios to resolve crisis

Highly unlikely due to solar system formation history

- *Scenario 1: Canonical impact + Earth-like composition impactor*
- *Scenario 2: Hit-and-run impact.*
- *Scenario 3: Fast-spinning Earth impact*
- *Scenario 4: Half-Earth impact.*
- *Scenario 5: General synestia impact with equilibration*
- *Scenario 6: Multiple impacts.*
- *Scenario 7: Canonical impact + Earth magma ocean*
- *Scenario 8: Canonical impact + equilibration*

Still impact-rich disk

Need extraction of angular momentum from system $L_{sys} \approx 1.4 L_{EM}$

Need significant extraction of angular momentum from system $L_{sys} \approx 2.5 L_{EM}$

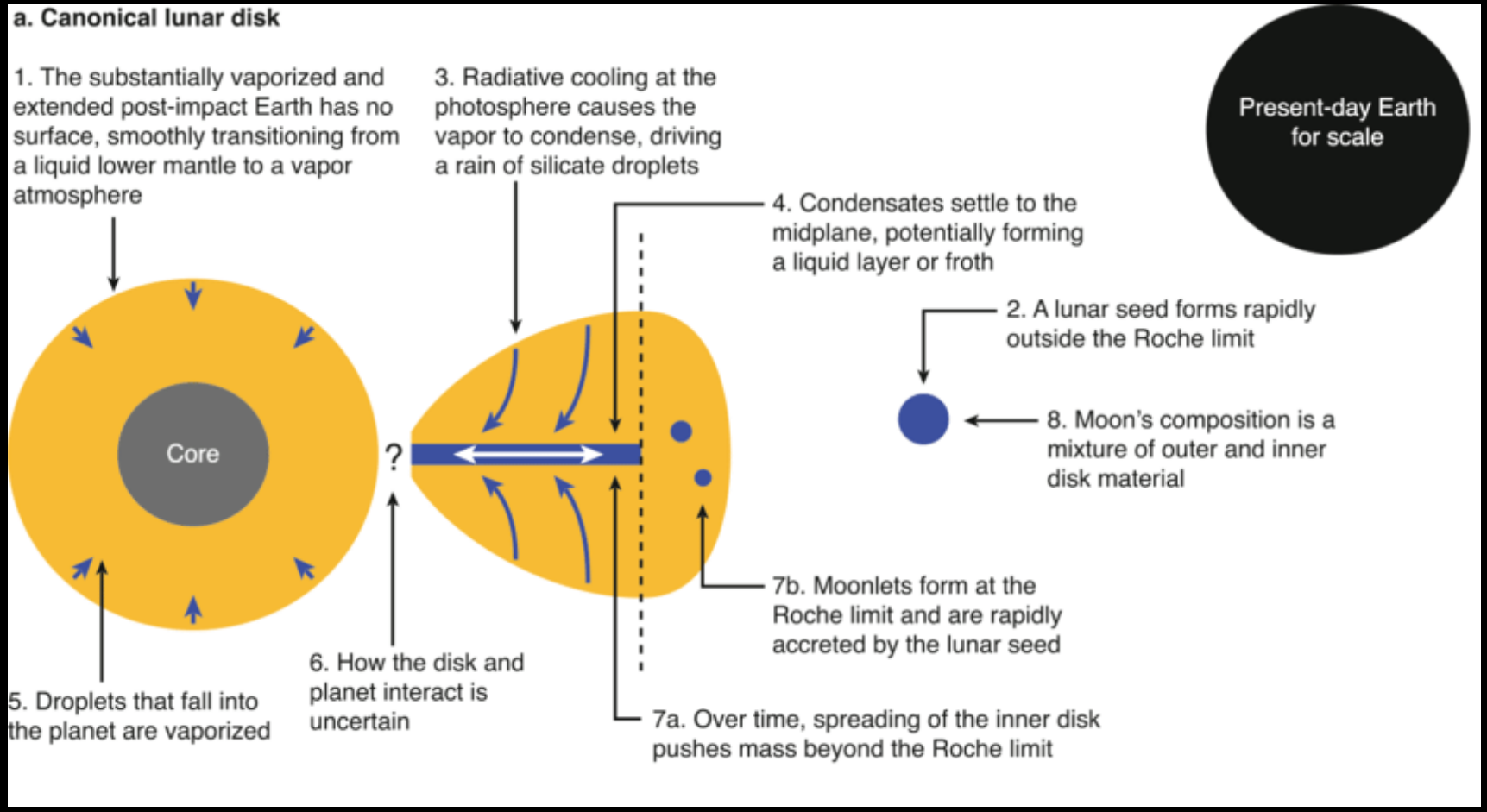
Dependent on impact history

High "success" rate associated with higher velocity impactors (unlikely)

Equation of state and numerical method needs to be further validated

Initial depth of magma ocean uncertain

Mixing after the impact

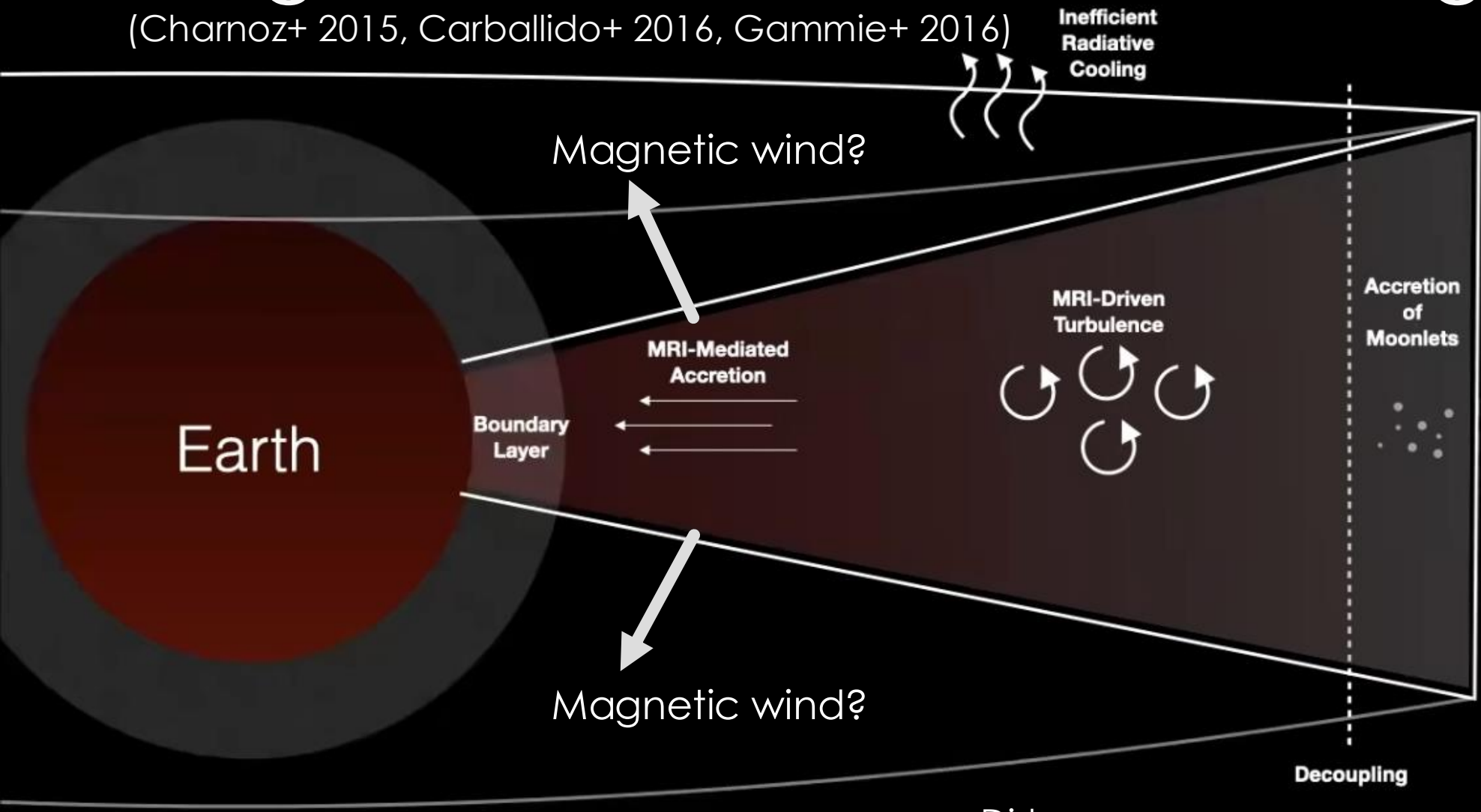


Lock et al. 2017

- Convective mixing between vapor components of disk and Earth (Pahlevan & Stevenson (2007)
- Assuming that the disk and planet interaction is effective ($t_{mix} \approx t_{spread}$)
- Mixing of Titanium requires disk to be very hot and remain hot for long time (Zhang et al. 2012)

Magnetic field –> increase mixing?

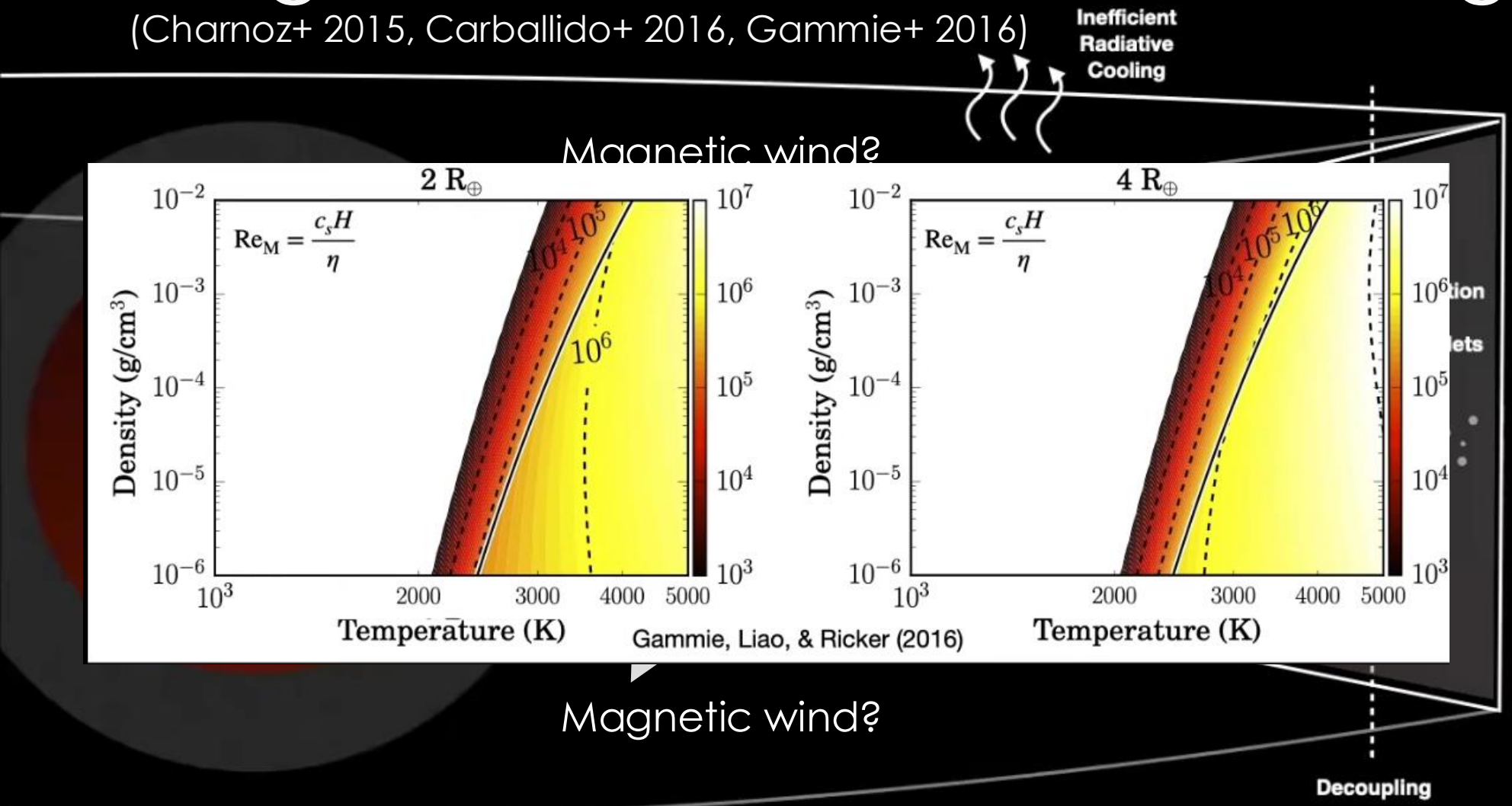
(Charnoz+ 2015, Carballido+ 2016, Gammie+ 2016)



- Disk $t_{heat} > t_{cool}$
  Turbulence and heating -> **increases t_{spread}**

Magnetic field –> increase mixing?

(Charnoz+ 2015, Carballido+ 2016, Gammie+ 2016)



Magnetic Reynolds number (Re_M) describes effective "coupling" of magnetic field

Activation of magnetorotational instability

- Disk $t_{heat} > t_{cool}$
  Turbulence and heating -> **increases t_{spread}**

Magnetorotational instability (MRI)

- Local linear instability that occurs in disks satisfying:

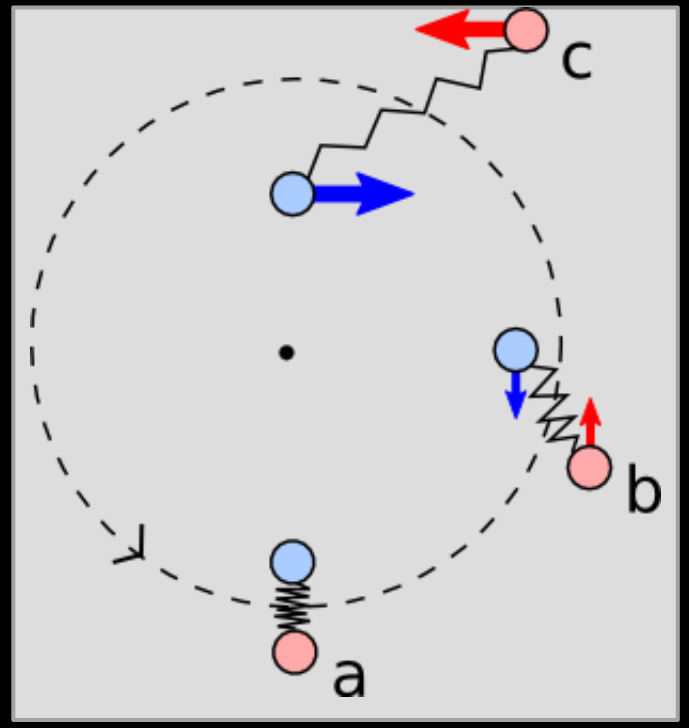
$$\frac{d\Omega^2}{d\ln R} < 0$$

$$k^2 \frac{B^2}{\rho} < -\frac{d\Omega^2}{d\ln R}$$

- Requires sufficiently ionized gas

$$\Lambda = \frac{B^2}{\rho\eta\Omega} > 1$$

- Leads to exponential growth of the field



Magnetorotational instability (MRI)

- Local linear instability that occurs in disks satisfying:

$$\frac{d\Omega^2}{d\ln R} < 0$$

$$k^2 \frac{B^2}{\rho} < -\frac{d\Omega^2}{d\ln R}$$

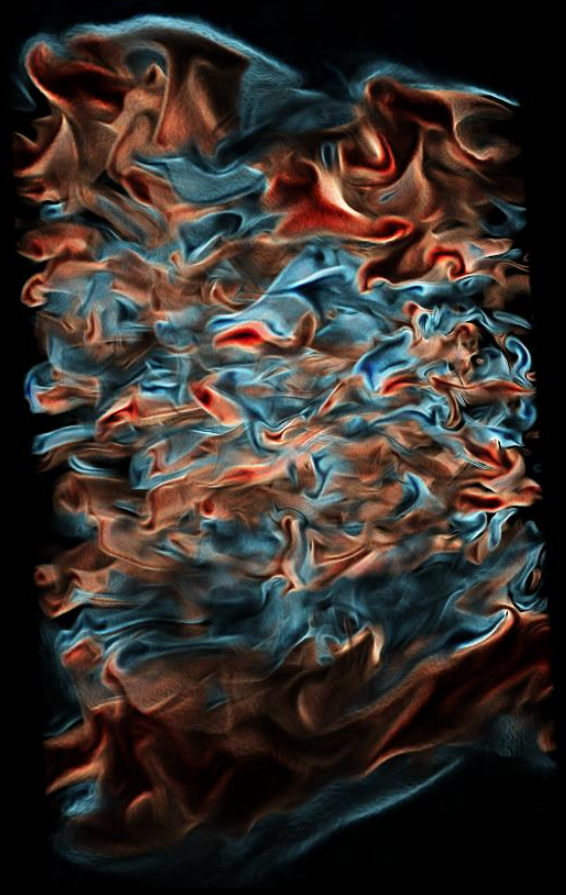
- Requires sufficiently ionized gas

$$\Lambda = \frac{B^2}{\rho\eta\Omega} > 1$$

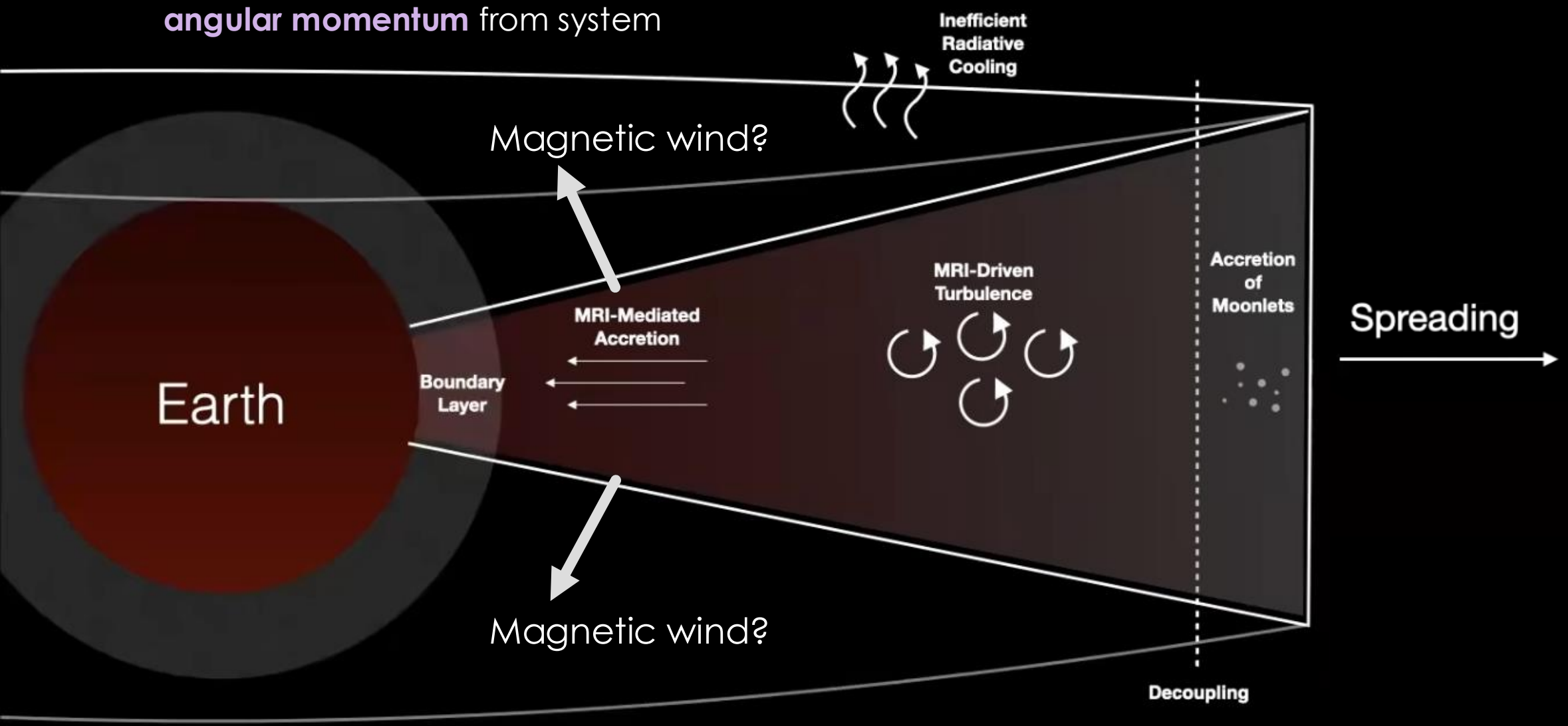
- Leads to exponential growth of the field

- The instability eventually breaks down into MHD turbulence
  - Self-sustaining process driven by non-linear dynamics

- Ensued turbulence acts to heat and increase the spreading of the protolunar disk

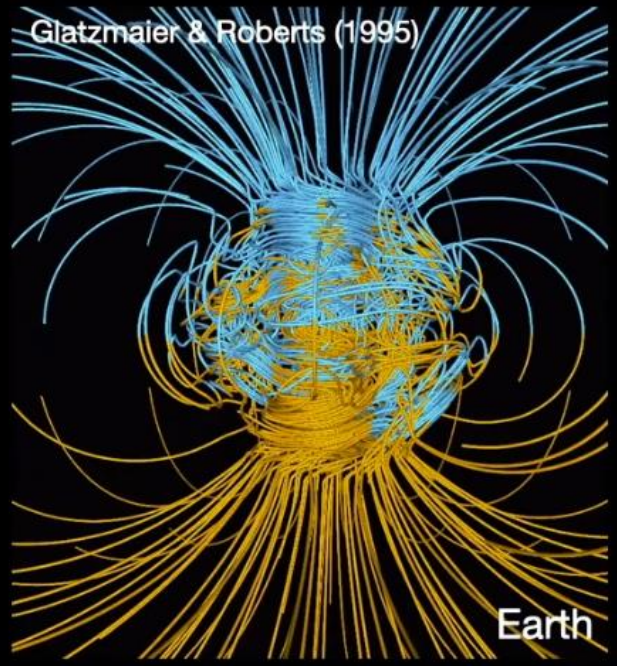


Magnetically driven outflow can **extract angular momentum** from system



Possible that decoupling occurs further out, increasing **disk edge to $\sim 10 R_E$**

Sources of magnetic field in the solar system

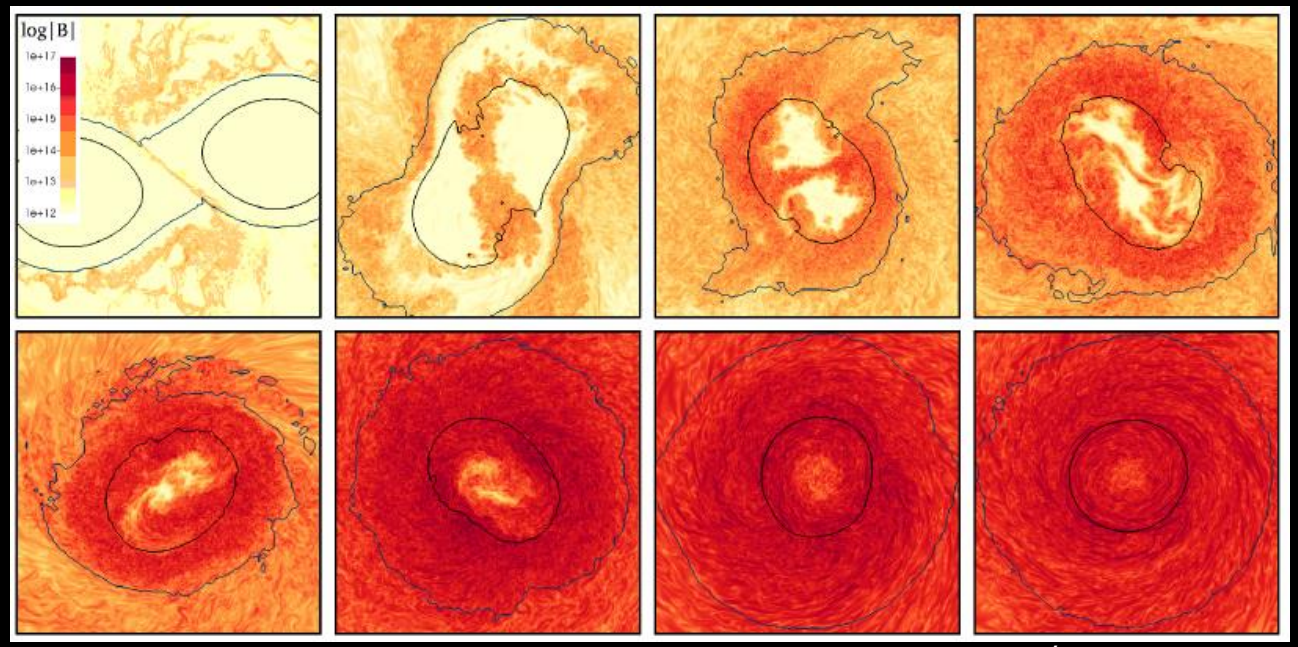


All solar system planets except Venus have been shown to have hosted or currently host a magnetic field.

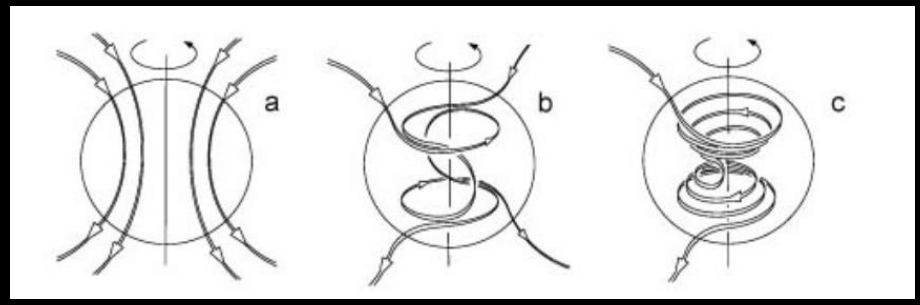
Meteoritic data suggest background solar magnetic field strength of around 0.05-0.5 G within the region
(Stevenson 2010, Cournede+ 2015, Weiss+ 2017)

Amplification processes of magnetic fields during giant impact

- Magneto-rotational instability in disk
- Turbulent amplification
  - Magnetic fields can amplify orders of magnitude during neutron star mergers (Palenzuela 2022)
- Magnetic winding
  - Linear growth of toroidal fields due to winding of radial fields.
  - Amplification will be based on the net radial magnetic flux in the disk after merger



(Palenzuela 2022)



A magnetized lunar-forming impact

(Mullen and Gammie 2020)

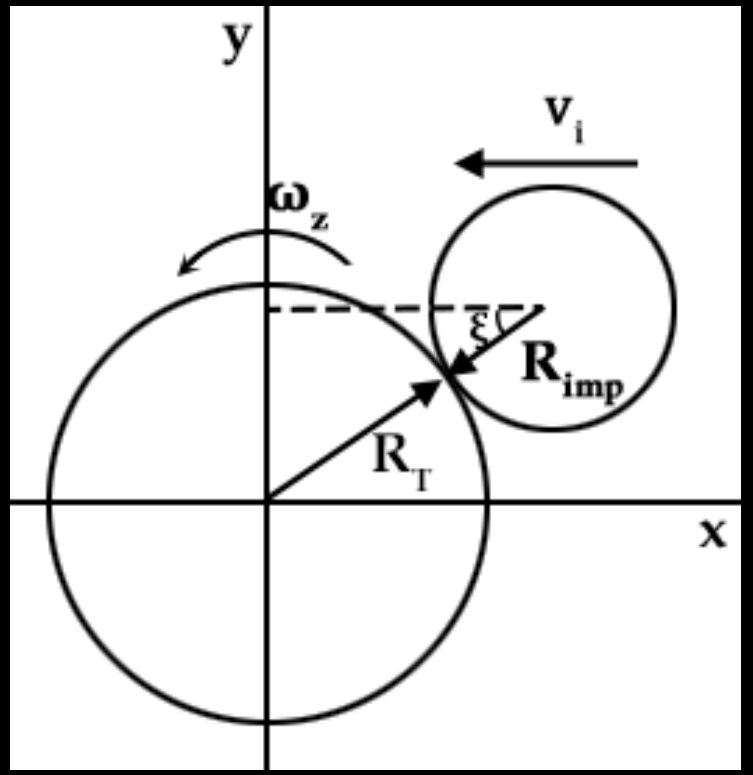
- Code: Athena++ Mesh
- Resolution

$$\Delta x = \{350, 175\}\, km$$

- Boundary condition = Outflowing
- Impact parameters

$$v_i = 1.2\, v_{esc}$$
$$\omega_z = 0$$
$$\xi = 35°$$
$$M_{target} = 0.9 M_E \qquad M_{impactor} = 0.2 M_E$$

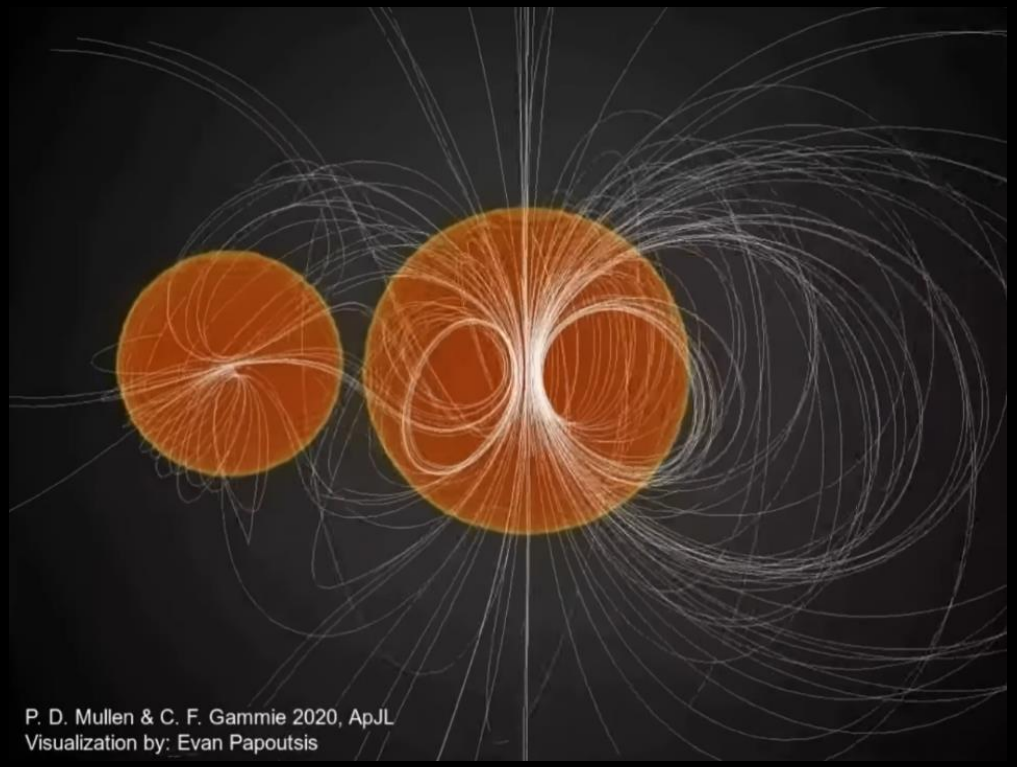
- No iron core
- Tillotson EOS for silicates



A magnetized lunar-forming impact

(Mullen and Gammie 2020)

- Ideal MHD model
- The initial dipole is characterized by the surface field strength at the magnetic poles, B_0.
  - $B_0 = \{B_{crit}\ , 10^{-9} B_{crit}\}$
- To resolve the fastest growing unstable mode of the MRI, the magnetic field needs to be stronger than B_{crit}
  - $B_{crit} \approx 1kG$! In their simulation

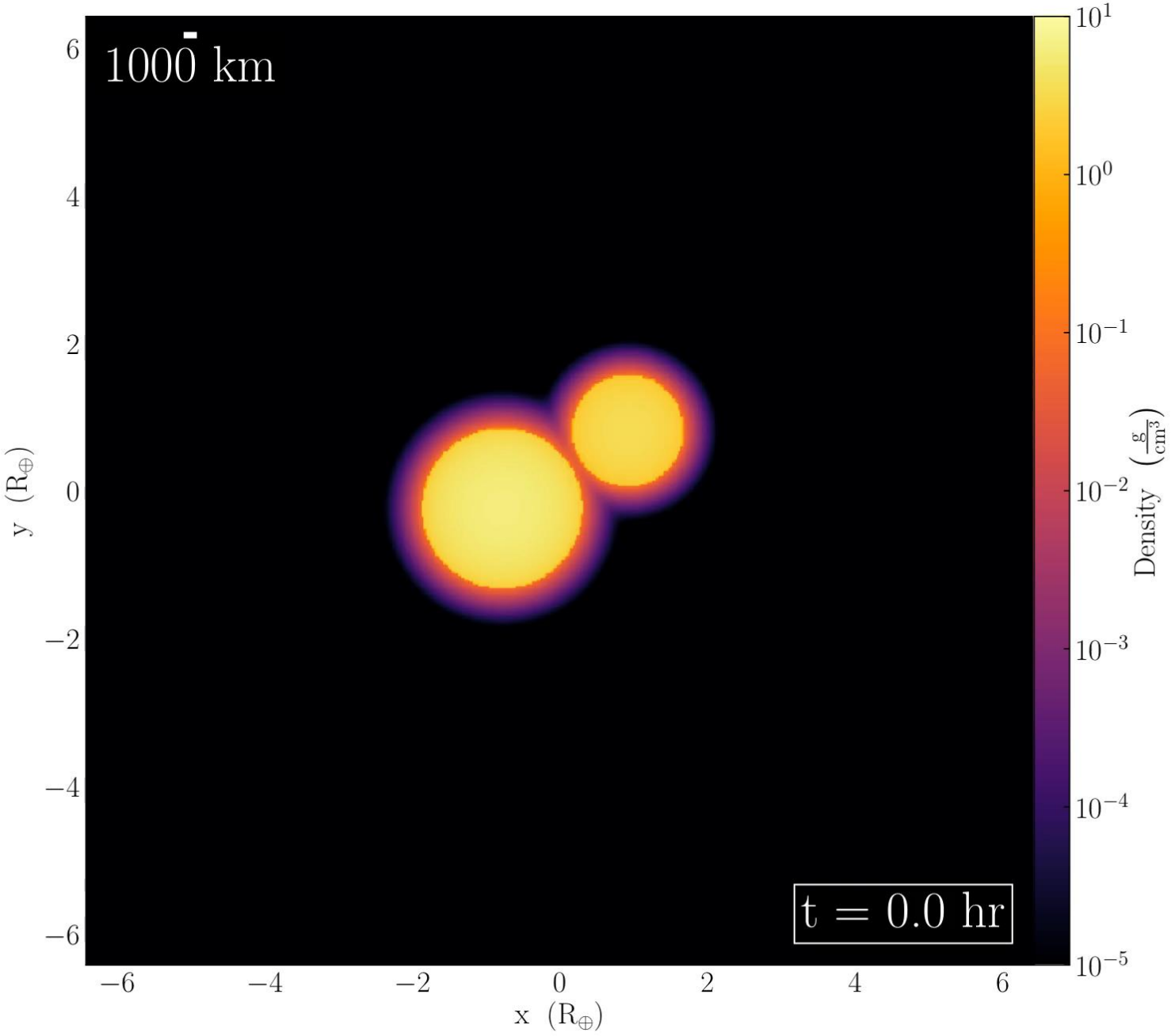


$$B_{\rm crit} \gtrsim 2 \times 10^3 \left(\frac{\Delta x}{250\ {\rm km}}\right) \left(\frac{\rho_{\rm midplane}}{10^{-3}\ {\rm g/cm^3}}\right)^{1/2} \left(\frac{R}{2R_\oplus}\right)^{-3/2} {\rm G}$$

At 48 hr post-impact, the boundary layer magnetic field at the midplane is ~8 kG

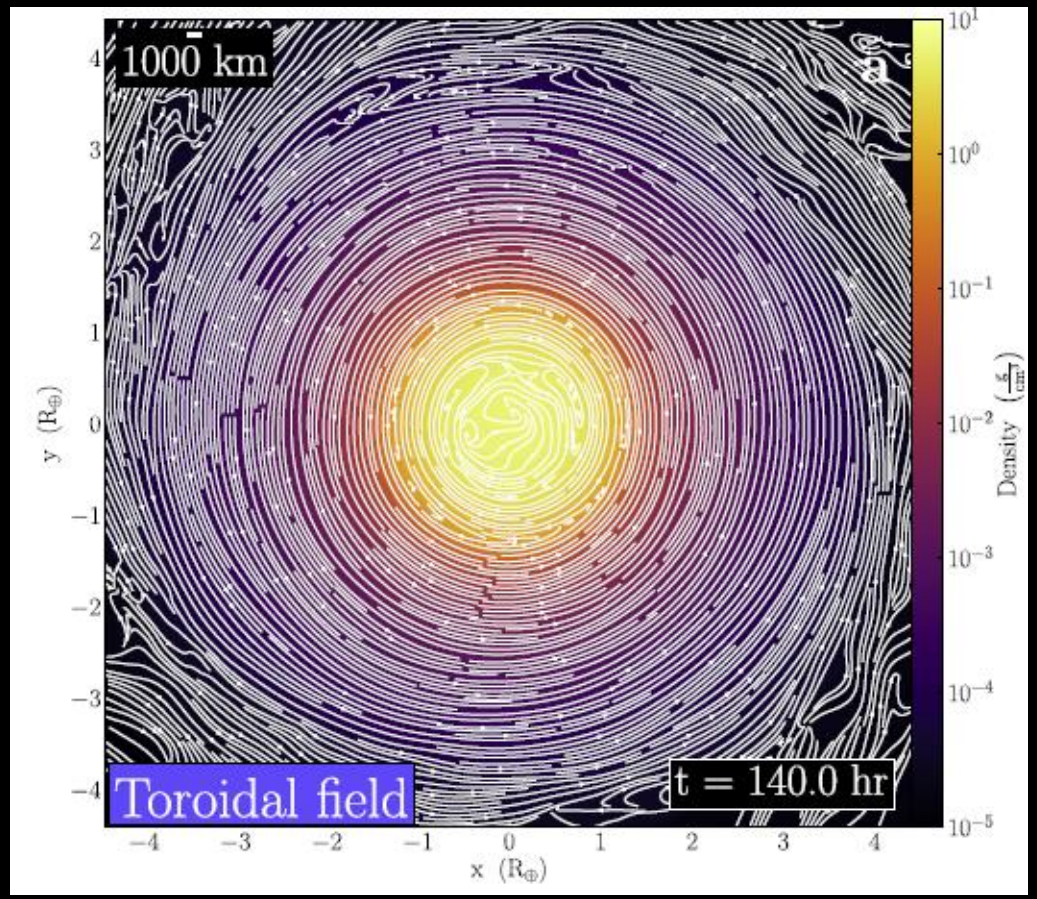
In the disk midplane, field strength declines with increasing radius, reaching ~350 G at $R \sim 4R_E$ so below the critical magnetic field strength for MRI

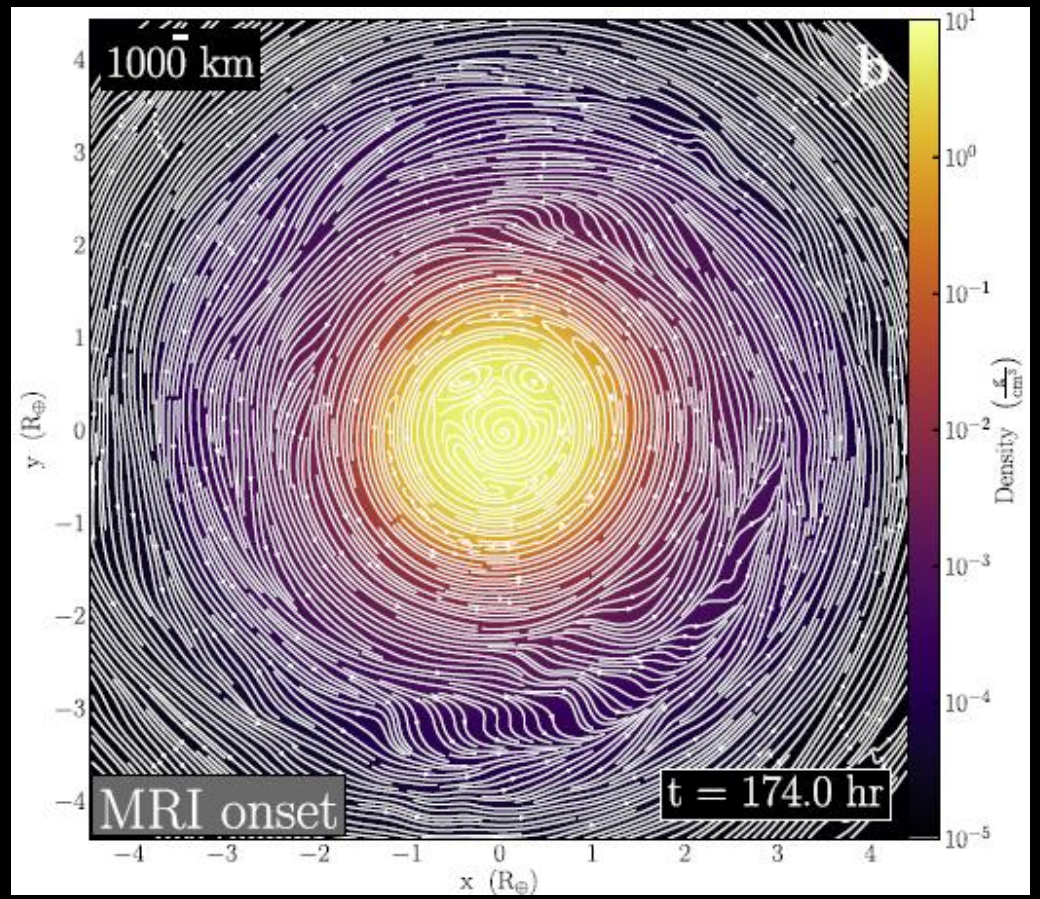
Higher resolution expected to result in higher levels of amplification due to turbulence during merger.



A magnetized lunar-forming impact

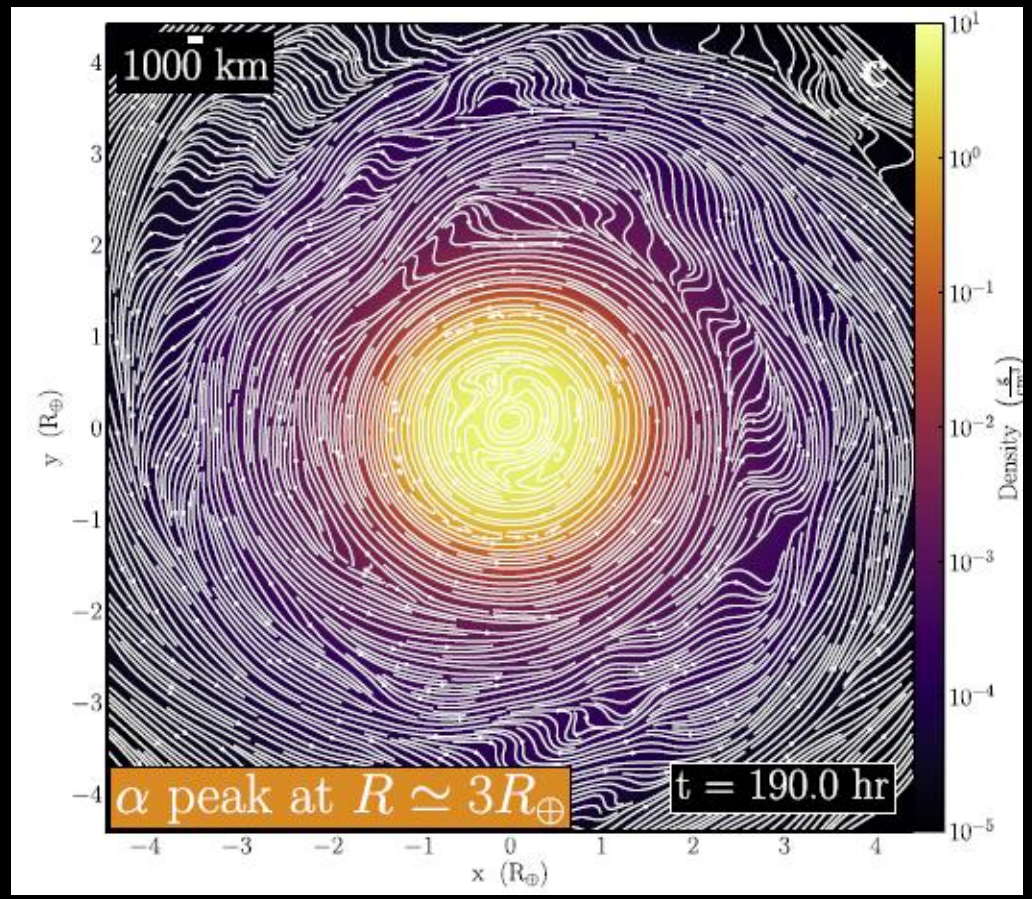
(Mullen and Gammie 2020)

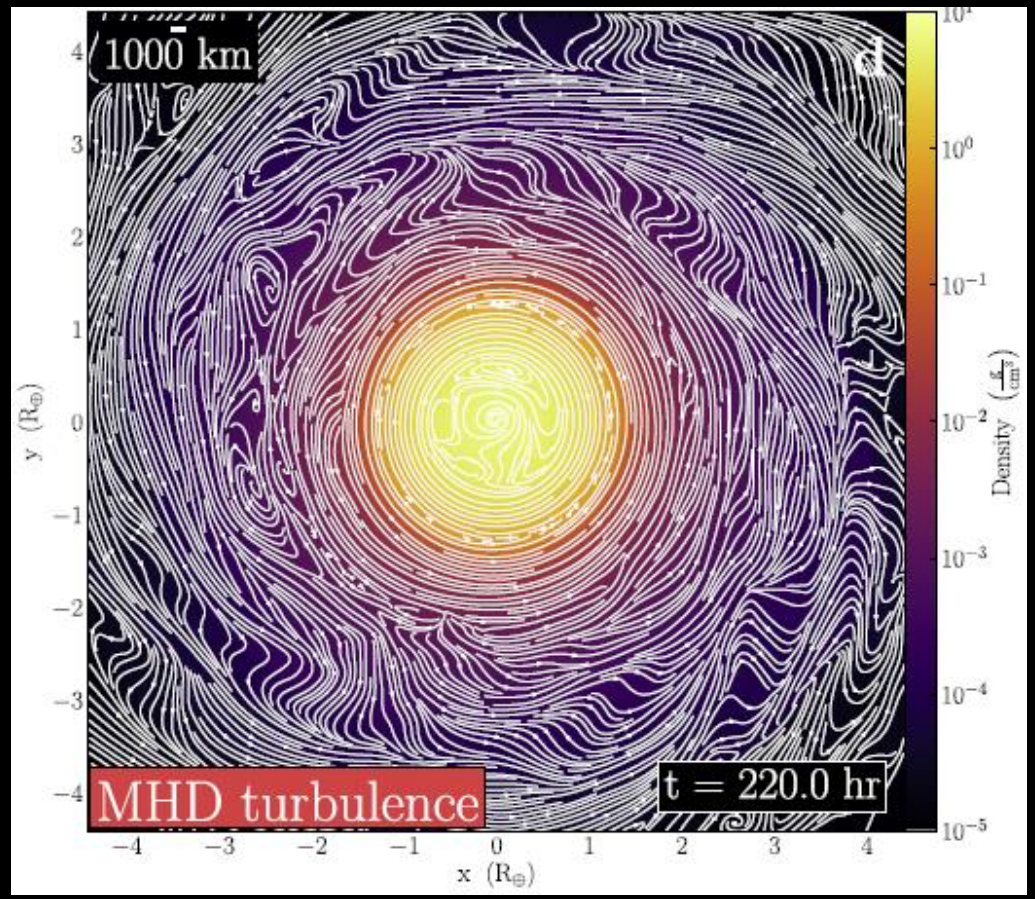




A magnetized lunar-forming impact

(Mullen and Gammie 2020)





A magnetized lunar-forming impact

(Mullen and Gammie 2020)

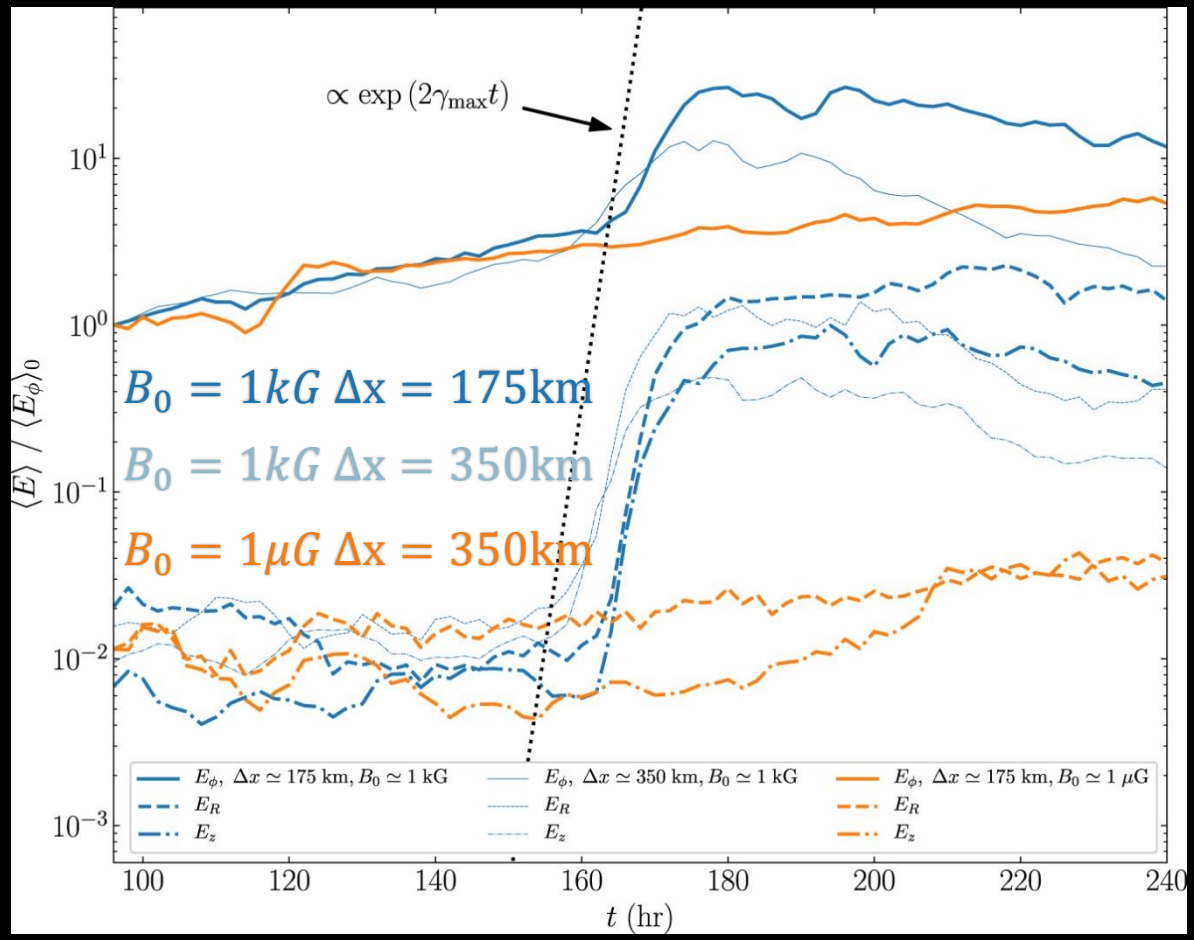
- Volume averaged magnetic energies over cylindrical annuli

$(R_{inner}, R_{outer}, h)_{\Delta x\,=\,175\ km} = (3R_E, 4R_E, 0.5R_E),$

$(R_{inner}, R_{outer}, h)_{\Delta x\,=\,350\ km} = (3.85R_E, 4.85R_E, 0.5R_E).$

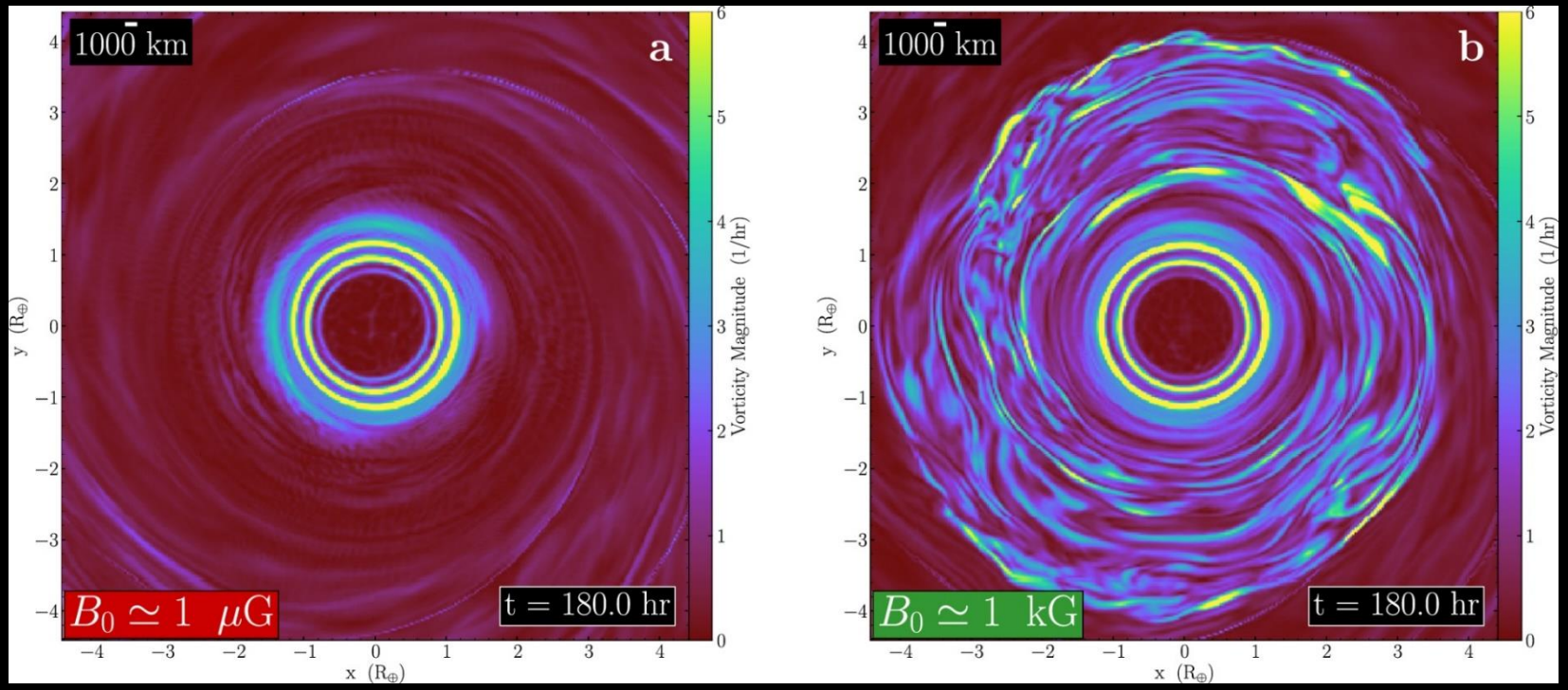
- Linear field growth of field due to winding
- Critical magnetic field reached in the protolunar disk
- Clear signature of exponential MRI growth at 174hrs

- Low initial magnetic field does not excite the MRI and only exhibit growth due to winding

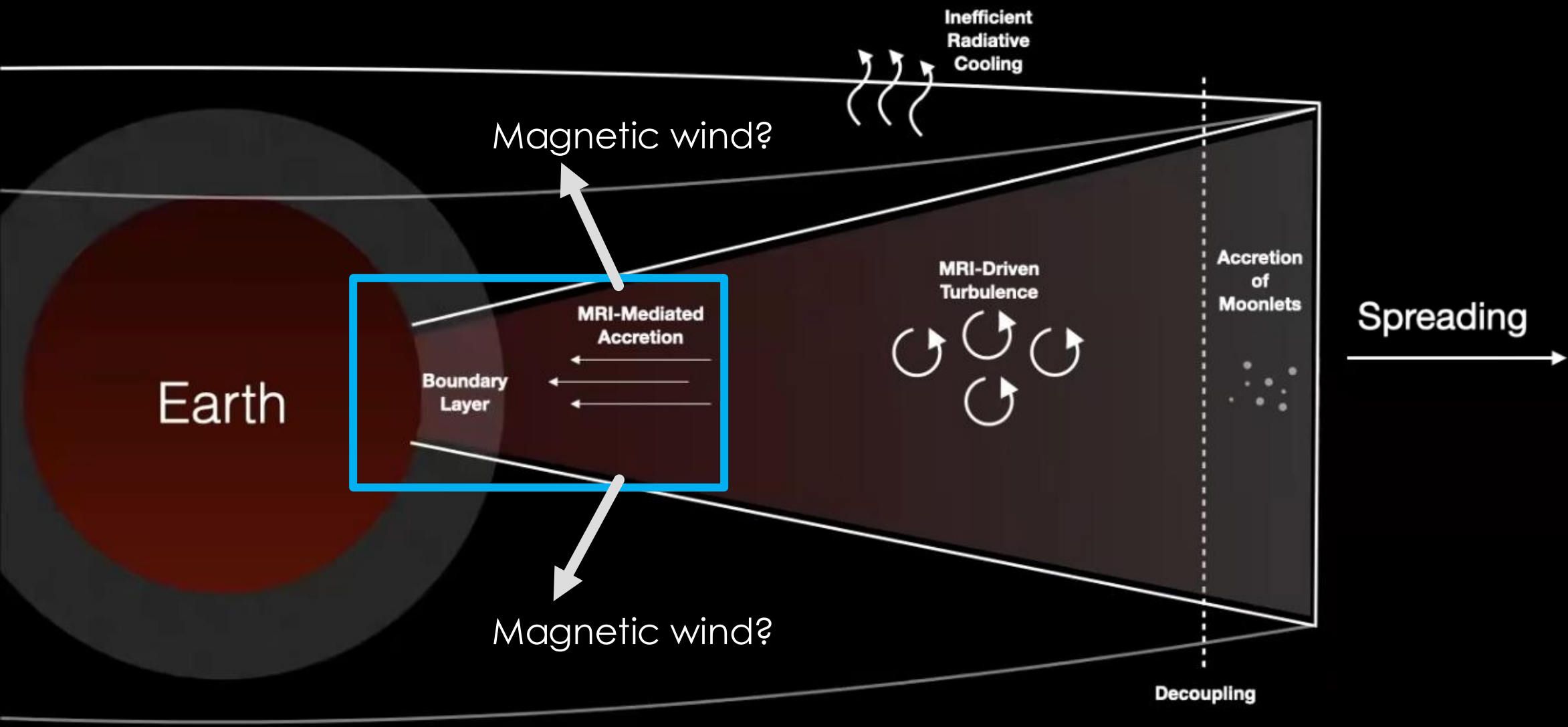


A magnetized lunar-forming impact

(Mullen and Gammie 2020)



- Spreading of protolunar disk go from $t_{spread} \sim 100\ years$ to $t_{spread} \sim \mathbf{100\ hours!}$
- Hastening the formation of the moon
- Disk material accretes onto proto-earth at a faster rate, potentially making moon formation less efficient
- Consequence might be that we need larger disk masses to form the Moon



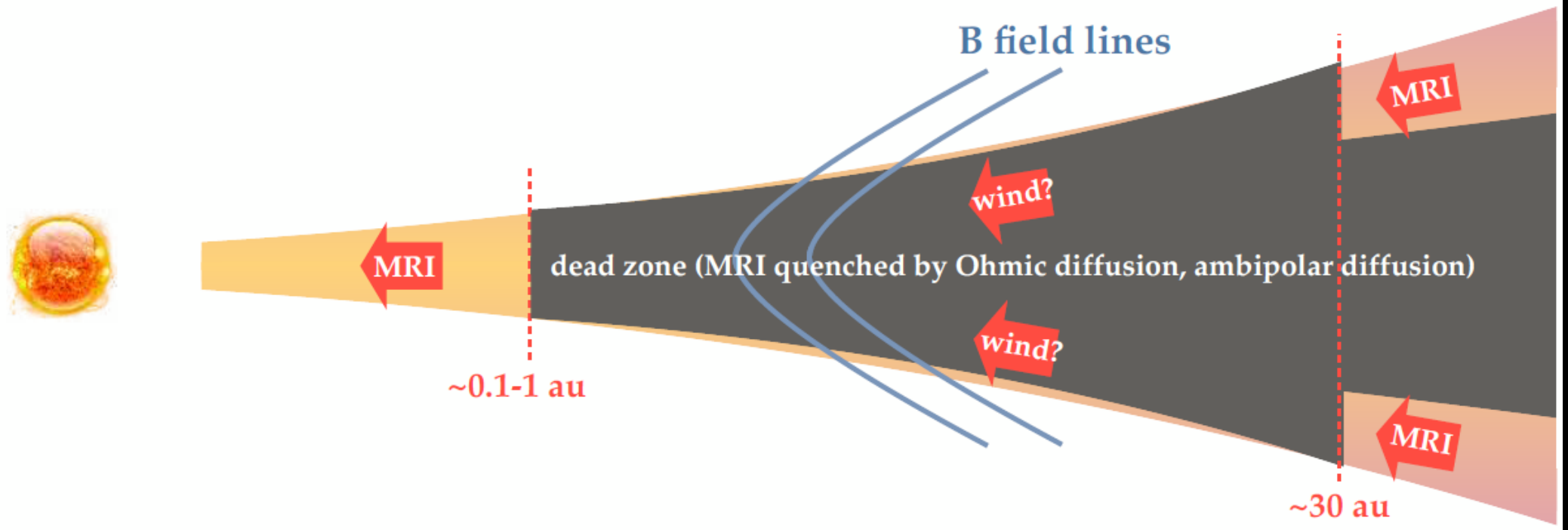
Mixing with proto-Earth will heavily depend on the boundary layer!!! $t_{mix} \approx t_{spread}$?

Things to consider

- Ideal MHD used! Resistive MHD effects can significantly quench the effect of MRI

MRI in protoplanetary disk

protoplanetary disks are in fact **poorly** ionized! (n_e / n < 10^{-13})



→ Ambipolar diffusion (ions-neutrals collisions) largely **quenches** MRI in the disk's surface layers, and partly in its outer parts

→ **Wind**-driven accretion if a vertical B field threads the disk

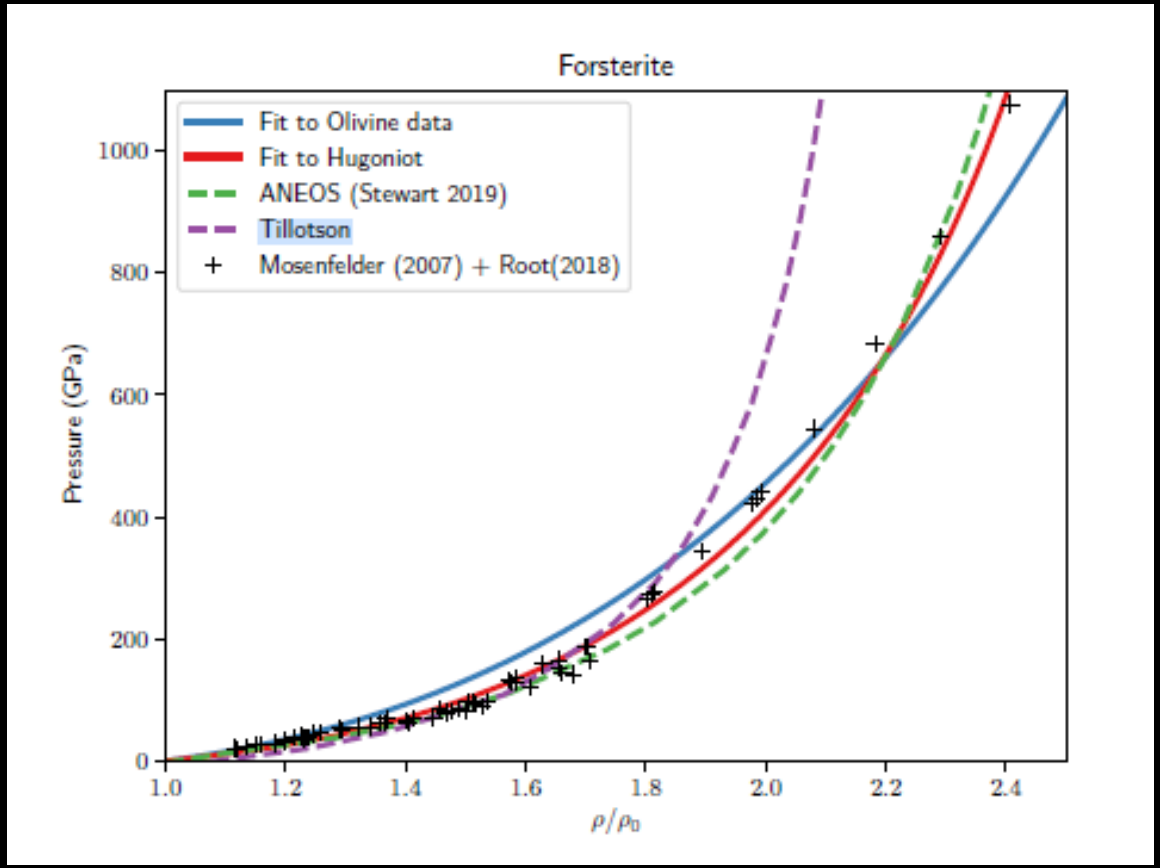
Bai 13, Simon+ 13…

Things to consider

- Ideal MHD used! Resistive MHD effects can significantly quench the effect of MRI
- Initial magnetic flux considerations
  - Unphysical initial magnetic field
    Can global magnetic fluxes change the MRI and the winding in the disk ? Magnetic outflows?
- Higher resolution might result in more turbulent amplification during merger.

Things to consider

- Ideal MHD used! Resistive MHD effects can significantly quench the effect of MRI
- Initial magnetic flux considerations
  - Unphysical initial magnetic field
    Can global magnetic fluxes change the MRI and the winding in the disk ? Magnetic outflows?
- Higher resolution might result in more turbulent amplification during merger.
- No iron core used for the two bodies
- Tillotson EOS used for silicates, which lack a proper vapor transition leading to erroneous decompression velocities (Stewart et al. 2019, Wissing et al. 2020)



Wissing & Hobbs 2020a

The role of magnetic fields during giant impacts in the solar system

- Spreading of protolunar disk go from $t_{spread} \sim 100\ years$ to $t_{spread} \sim \mathbf{100\ hours!}$
- **Hastening the formation of the moon**

- Potentially capture moon formation in the same dynamical simulation?

- Possible that magnetically driven outflow can **extract angular momentum** from system

- Disk material accretes onto proto-earth at a faster rate, **potentially making moon formation less efficient, requires more massive disk**

- If $t_{mix} \approx t_{spread}$ we could potentially mix disk and planet rapidly (depends on boundary layer)

- Possible that decoupling occurs further out, increasing **disk edge to $\sim 10 R_E$**

- **Uncertain and many things left to consider!**